

[GETNET]Upgrade Oracle Database 18c to 19c – Linux

1 – Ações antes da Janela (não geram indisponibilidade).

Criação da estrutura de diretórios 19c e db_profile (oracle):

```
[oracle@upgrade db]$ mkdir -p /u01/app/19/db/
[oracle@upgrade db]$ chmod 775 /u01/app/19/db/
[oracle@upgrade db]$ chmod 775 /u01/app/19/
[oracle@upgrade db]$ chown oracle:oinstall /u01/app/19/db/

[oracle@upgrade ~]$ cat db_profile_19c
export ORACLE_BASE=/u01/app/19/
export ORACLE_HOME=/u01/app/19/db/
export ORACLE_SID=ORAEX
export PATH=/u01/app/19/db/bin:/u01/app/19/db/bin:/usr/bin:/bin:/usr/sbin:/sbin
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"
```

Criação da estrutura de diretórios 19c e grid_profile (grid):

```
[grid@upgrade ~]$ mkdir -p /u01/app/19/grid/
[grid@upgrade ~]$ chmod 775 /u01/app/19/grid/
[grid@upgrade ~]$ chown grid:oinstall /u01/app/19/grid/

[grid@upgrade ~]$ cat grid_profile_19c
export ORACLE_BASE=/u01/app/19/
export ORACLE_HOME=/u01/app/19/grid/
export ORACLE_SID=+ASM
export PATH=/u01/app/19/grid/bin:/usr/bin:/bin:/usr/sbin:/sbin
```

Descompactar os binários do grid:

```
[grid@upgrade grid]$ unzip LINUX.X64_193000_grid_home.zip
```

Descompactar os binários do oracle:

```
[oracle@upgrade db]$ unzip LINUX.X64_193000_db_home.zip
```

Validação do Banco logo no inicio que iniciar a janela de GMUD:

A ideia é gerar evidências do status do Banco de Dados antes do upgrade:

```
SQL> col host_name format a20
SET LINES 200
select UPPER(instance_name) ins_name, host_name, database_status, version, status, to_char(startup_time,'dd-mm-yyyy hh24:mi') Started, login
s
from v$instance
/
```

INS_NAME	HOST_NAME	DATABASE_STATUS	VERSION	STATUS	STARTED	LOGINS
ORAEX	upgrade.localdomain	ACTIVE	18.0.0.0.0	OPEN	08-03-2021 18:59	ALLOWED

```
SQL> col HOST_NAME format a30
set lines 230
col COMP_NAME format a55
set pages 900
col DESCRIPTION format a65
col ACTION_TIME format a29
col COMMENTS format a40
col ACTION format a9
col NAMESPACE format a9
col ID format 99999999
```

```
SQL> select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv$instance, v$database;
```

INSTANCE_NAME	HOST_NAME	VERSION	STATUS	OPEN_MODE	DATABASE_ROLE	SWITCHOVER_STAT
ORAEX	upgrade.localdomain	18.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED

```
SQL> select comp_name, version, status, modified from dba_registry;
```

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Database Catalog Views	18.0.0.0.0	VALID	05-MAR-2021 21:56:51
Oracle Database Packages and Types	18.0.0.0.0	VALID	05-MAR-2021 21:56:51
Oracle Real Application Clusters	18.0.0.0.0	OPTION OFF	07-FEB-2018 21:05:23
JServer JAVA Virtual Machine	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle XDK	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Database Java Packages	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
OLAP Analytic Workspace	18.0.0.0.0	VALID	05-MAR-2021 21:56:55
Oracle XML Database	18.0.0.0.0	VALID	05-MAR-2021 21:56:53
Oracle Workspace Manager	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Text	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Multimedia	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Spatial	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle OLAP API	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle Label Security	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle Database Vault	18.0.0.0.0	VALID	05-MAR-2021 21:56:56

15 rows selected.

```
SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;
```

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
28090523	APPLY	SUCCESS	05-MAR-21 09.55.24.957775 PM	Database Release Update : 18.3.0.0.180717 (28090523)
27923415	APPLY	SUCCESS	05-MAR-21 09.55.24.963174 PM	OJVM RELEASE UPDATE: 18.3.0.0.180717 (27923415)

```
SQL> set lines 180
set pages 40000
col owner for a30
select owner, count(*) from dba_objects
where status = 'INVALID'
group by owner
/
```

no rows selected

Validações com user grid;

```
[grid@upgrade ~]$ . grid_profile
[grid@upgrade ~]$ srvctl config asm
ASM home: <CRS home>
Password file: +DATA_ORAEX/orapwasm
Backup of Password file:
ASM listener: LISTENER
Spfile: +DATA_ORAEX/ASM/ASMPARAMETERFILE/registry.253.1066424151
ASM diskgroup discovery string: /dev/oracleasm/disks/*
```

3- Ações durante a janela da GMUD (inicio da indisponibilidade)

Criação do PFILE e validação do Timezone_file:

```
SQL> create pfile='/u02/midia/initDB_18c.ora' from spfile;
```

File created.

```
SQL> select * from v$timezone_file;
```

FILENAME	VERSION	CON_ID
timezlrg_31.dat	31	0

Parar do Grid:

```

[grid@upgrade ~]$ . grid_profile_19c
[grid@upgrade ~]$ srvctl config database
ORAEX
[grid@upgrade ~]$ srvctl config database -d ORAEX
Database unique name: ORAEX
Database name: ORAEX
Oracle home: /u01/app/18/db
Oracle user: oracle
Spfile: +DATA_ORAEX/ORAEX/PARAMETERFILE/spfile.266.1066427785
Password file:
Domain:
Start options: open
Stop options: immediate
Database role: PRIMARY
Management policy: AUTOMATIC
Disk Groups: DATA_ORAEX
Services:
OSDBA group:
OSOPER group:
Database instance: ORAEX

[grid@upgrade ~]$ srvctl status database -d ORAEX
Database is running.
[grid@upgrade ~]$ srvctl stop database -d ORAEX

[grid@upgrade ~]$ srvctl status listener -l LISTENER
Listener LISTENER is enabled
Listener LISTENER is running on node(s): upgrade
[grid@upgrade ~]$ srvctl stop listener -l LISTENER

[grid@upgrade ~]$ crsctl stop has
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'upgrade'
CRS-2673: Attempting to stop 'ora.ARCH_ORAEX.dg' on 'upgrade'
CRS-2673: Attempting to stop 'ora.DATA_AUX_ORAEX.dg' on 'upgrade'
CRS-2673: Attempting to stop 'ora.DATA_ORAEX.dg' on 'upgrade'
CRS-2677: Stop of 'ora.ARCH_ORAEX.dg' on 'upgrade' succeeded
CRS-2677: Stop of 'ora.DATA_AUX_ORAEX.dg' on 'upgrade' succeeded
CRS-2677: Stop of 'ora.DATA_ORAEX.dg' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.asm' on 'upgrade'
CRS-2677: Stop of 'ora.asm' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'upgrade'
CRS-2677: Stop of 'ora.evmd' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'upgrade'
CRS-2677: Stop of 'ora.cssd' on 'upgrade' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'upgrade' has completed
CRS-4133: Oracle High Availability Services has been stopped.

```

Pré Upgrade:

NOTA: no meu caso, ignoro o Swap por falta de robustez no meu host:

```

[grid@upgrade ~]$ /u01/app/19/grid/runcluvfy.sh stage -pre hacfg

Verifying Physical Memory ...PASSED
Verifying Available Physical Memory ...PASSED
Verifying Swap Size ...FAILED (PRVF-7573)
Verifying Free Space: upgrade:/usr,upgrade:/var,upgrade:/etc,upgrade:/sbin,upgrade:/tmp ...PASSED
Verifying User Existence: grid ...
  Verifying Users With Same UID: 54322 ...PASSED
Verifying User Existence: grid ...PASSED
Verifying Group Existence: dba ...PASSED
Verifying Group Membership: dba ...PASSED
Verifying Run Level ...PASSED
Verifying Architecture ...PASSED
Verifying OS Kernel Version ...PASSED
Verifying OS Kernel Parameter: semmsl ...PASSED
Verifying OS Kernel Parameter: semmns ...PASSED
Verifying OS Kernel Parameter: semopm ...PASSED
Verifying OS Kernel Parameter: semmni ...PASSED
Verifying OS Kernel Parameter: shmmax ...PASSED
Verifying OS Kernel Parameter: shmmni ...PASSED

```

```
verrifying OS Kernel Parameter: snmail ...PASSED
Verifying OS Kernel Parameter: file-max ...PASSED
Verifying OS Kernel Parameter: ip_local_port_range ...PASSED
Verifying OS Kernel Parameter: rmem_default ...PASSED
Verifying OS Kernel Parameter: rmem_max ...PASSED
Verifying OS Kernel Parameter: wmem_default ...PASSED
Verifying OS Kernel Parameter: wmem_max ...PASSED
Verifying OS Kernel Parameter: aio-max-nr ...PASSED
Verifying Package: kmod-20-21 (x86_64) ...PASSED
Verifying Package: kmod-libs-20-21 (x86_64) ...PASSED
Verifying Package: binutils-2.23.52.0.1 ...PASSED
Verifying Package: compat-libcap1-1.10 ...PASSED
Verifying Package: libgcc-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-devel-4.8.2 (x86_64) ...PASSED
Verifying Package: sysstat-10.1.5 ...PASSED
Verifying Package: gcc-c++-4.8.2 ...PASSED
Verifying Package: ksh ...PASSED
Verifying Package: make-3.82 ...PASSED
Verifying Package: glibc-2.17 (x86_64) ...PASSED
Verifying Package: glibc-devel-2.17 (x86_64) ...PASSED
Verifying Package: libaio-0.3.109 (x86_64) ...PASSED
Verifying Package: libaio-devel-0.3.109 (x86_64) ...PASSED
Verifying Package: nfs-utils-1.2.3-15 ...PASSED
Verifying Package: smartmontools-6.2-4 ...PASSED
Verifying Package: net-tools-2.0-0.17 ...PASSED
Verifying Package: compat-libstdc++-33-3.2.3 (x86_64) ...PASSED
Verifying Package: libxcb-1.11 (x86_64) ...PASSED
Verifying Package: libX11-1.6.3 (x86_64) ...PASSED
Verifying Package: libXau-1.0.8 (x86_64) ...PASSED
Verifying Package: libXi-1.7.4 (x86_64) ...PASSED
Verifying Package: libXtst-1.2.2 (x86_64) ...PASSED
Verifying Users With Same UID: 0 ...PASSED
Verifying Current Group ID ...PASSED
Verifying Root user consistency ...PASSED
```

Pre-check for Oracle Restart configuration was unsuccessful.

Failures were encountered during execution of CVU verification request "stage -pre hacfg".

Verifying Swap Size ...FAILED

upgrade: PRVF-7573 : Sufficient swap size is not available on node "upgrade"
[Required = 7.6382GB (8009236.0KB) ; Found = 3GB (3145724.0KB)]

CVU operation performed: stage -pre hacfg
Date: Mar 9, 2021 3:55:59 PM
CVU home: /u01/app/19/grid/
User: grid

Configuração da grade:

```
[grid@upgrade ~]$ . grid_profile_19c
[grid@upgrade ~]$ cd $ORACLE_HOME
[grid@upgrade ~]$ ./gridSetup.sh
```

Oracle Grid Infrastructure 19c Installer - Step 1 of 9

Select Configuration Option

Configuration Option

Cluster Configuration

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select an option to configure the software. The wizard will register the home in the central inventory and then perform the selected configuration.

☐ Configure Oracle Grid Infrastructure for a New Cluster

☐ Configure Oracle Grid Infrastructure for a Standalone Server (Oracle Restart)

☒ Upgrade Oracle Grid Infrastructure

☐ Set Up Software Only

Help

< Back

Next >

Install

Cancel

Marcamos a opção de Upgrade Oracle Grid Infrastructure

Oracle Grid Infrastructure 19c Installer - Step 2 of 8

Specify Management Options

Configuration Option

Management Options

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

You can configure to have this instance of Oracle Grid Infrastructure and Oracle Automatic Storage Management to be managed by Enterprise Manager Cloud Control. Specify the details of the Cloud Control configuration to perform the registration.

☐ Register with Enterprise Manager (EM) Cloud Control

O~~M~~S host:

O~~M~~S port:

EM Admin User Name:

EM Admin Password:

Help

< Back

Next >

Submit

Cancel

Padrão da Getnet é não configura EM visto que temos o OEM

Core Assets/Documentation/GETNET - DBA
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Oracle Grid Infrastructure 19c Installer - Step 3 of 8

Specify Installation Location

19c ORACLE
Grid Infrastructure

[Configuration Option](#)
[Management Options](#)
[Installation Location](#)
[Root script execution](#)
Prerequisite Checks
Summary
Install Product
Finish

Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.

Oracle base:

This software directory is the Oracle Grid Infrastructure home directory.

Software location: /u01/app/19/grid

Seguimos a estrutura de pastas do grid_profile. Eu estou seguindo de acordo com os hosts Linux que encontrei na GETNET. Mudando de /u01/app/18/grid para /u01/app/19/grid.

Oracle Grid Infrastructure 19c Installer - Step 4 of 8

Root script execution configuration

19c ORACLE
Grid Infrastructure

[Configuration Option](#)
[Management Options](#)
[Installation Location](#)
[Root script execution](#)
[Prerequisite Checks](#)
Summary
Install Product
Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

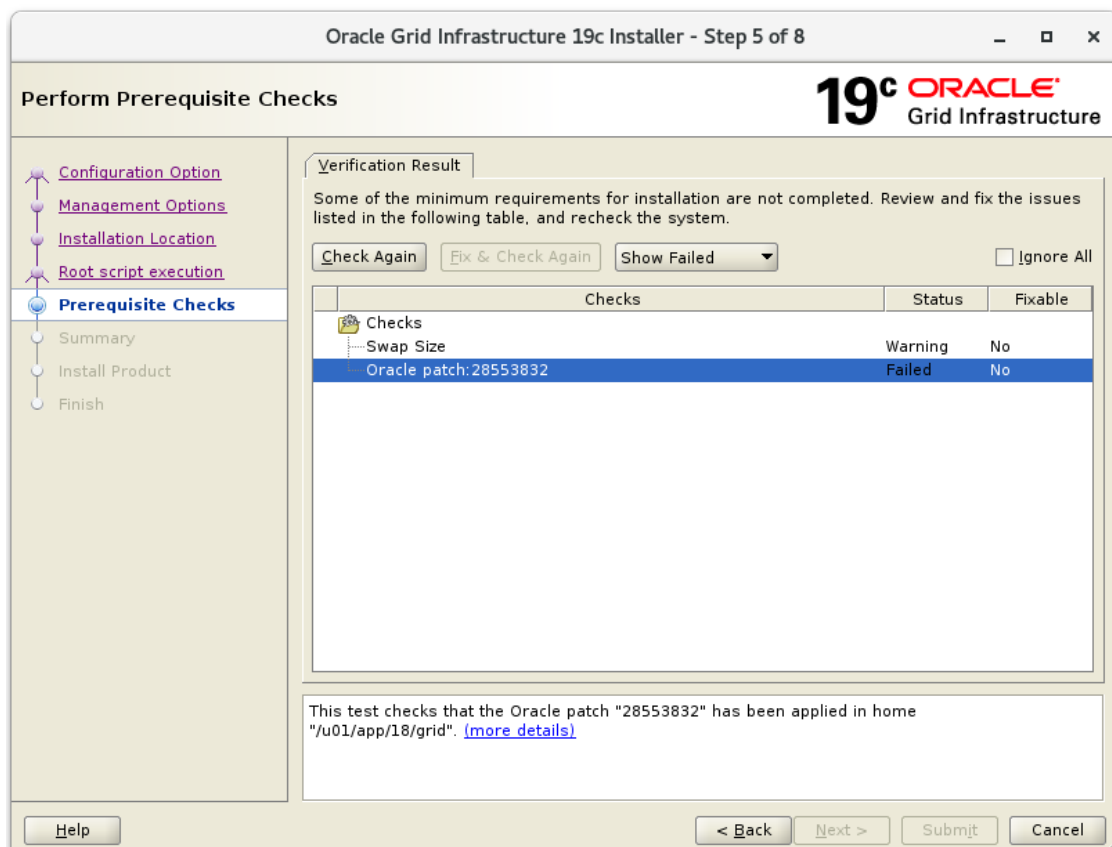
☐ Use sudo

Program path :

User name :

Password :

Como Best Practice ORAEX, não estou configurando os scripts root para serem executados automaticamente.
NOTA: Conforme previsto, não foi possível o upgrade direto do 18.3:



Sendo assim, cancelei a instalação e e apliquei o último PSU do 18c :

Aplicando o PSU no 18c:

Ajustando o OPatch do Grid:

```
[root@upgrade db]# cd /u02/midia/
[root@upgrade midia]# chown grid:oinstall p6880880_121010_Linux-x86-64.zip
[root@upgrade midia]# chmod 775 p6880880_121010_Linux-x86-64.zip
[root@upgrade midia]# . /home/grid/grid_profile
[root@upgrade midia]# cd $ORACLE_HOME
[root@upgrade grid]# cp /u02/midia/p6880880_121010_Linux-x86-64.zip .
[root@upgrade grid]# mv OPatch OPatch_old
[root@upgrade grid]# unzip p6880880_121010_Linux-x86-64.zip
...
[root@upgrade grid]# chown -R grid:oinstall OPatch
```

Ajustando o OPatch do Oracle:

```
[oracle@upgrade ~]$ . db_profile
[oracle@upgrade ~]$ cd $ORACLE_HOME
[oracle@upgrade db]$ cp /u02/midia/p6880880_121010_Linux-x86-64.zip .
[oracle@upgrade db]$ pwd
/u01/app/18/db
[oracle@upgrade db]$ mv OPatch OPatch_old
[oracle@upgrade db]$ unzip p6880880_121010_Linux-x86-64.zip
...
```

Como eu havia baixado o Grid e o Banco para fazer o upgrade, subi novamente os componentes para aplicar o último PSU disponível para 18c:

```
[root@upgrade OPatch]# ./opatchauto apply /u02/midia/32226219/
```

OPatchauto session is initiated at Tue Mar 9 17:06:26 2021

System initialization log file is /u01/app/18/grid/cfgtoollogs/opatchautodb/systemconfig2021-03-09_05-06-30PM.log.

Session log file is /u01/app/18/grid/cfgtoollogs/opatchauto/opatchauto2021-03-09_05-06-39PM.log

The id for this session is HXJG

Executing OPatch prereq operations to verify patch applicability on home /u01/app/18/db
Patch applicability verified successfully on home /u01/app/18/db

Executing patch validation checks on home /u01/app/18/db
Patch validation checks successfully completed on home /u01/app/18/db

Verifying SQL patch applicability on home /u01/app/18/db
SQL patch applicability verified successfully on home /u01/app/18/db

Executing OPatch prereq operations to verify patch applicability on home /u01/app/18/grid
Patch applicability verified successfully on home /u01/app/18/grid

Executing patch validation checks on home /u01/app/18/grid
Patch validation checks successfully completed on home /u01/app/18/grid

Preparing to bring down database service on home /u01/app/18/db
Successfully prepared home /u01/app/18/db to bring down database service

Bringing down database service on home /u01/app/18/db
Following database has been stopped and will be restarted later during the session: oraex
Database service successfully brought down on home /u01/app/18/db

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/18/grid
Prepatch operation log file location: /u01/app/18/crsdata/upgrade/crsconfig/hapatch_2021-03-09_05-08-41PM.log
CRS service brought down successfully on home /u01/app/18/grid

Start applying binary patch on home /u01/app/18/db
Binary patch applied successfully on home /u01/app/18/db

Start applying binary patch on home /u01/app/18/grid
Binary patch applied successfully on home /u01/app/18/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/18/grid
Postpatch operation log file location: /u01/app/18/crsdata/upgrade/crsconfig/hapatch_2021-03-09_05-33-00PM.log
CRS service started successfully on home /u01/app/18/grid

Starting database service on home /u01/app/18/db
Database service successfully started on home /u01/app/18/db

Preparing home /u01/app/18/db after database service restarted
No step execution required.....

Trying to apply SQL patch on home /u01/app/18/db
SQL patch applied successfully on home /u01/app/18/db

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:upgrade
SIDB Home:/u01/app/18/db
Version:18.0.0.0
Summary:

==Following patches were SKIPPED:

Patch: /u02/midia/32226219/32216944
Reason: This patch is not applicable to this specified target type - "oracle_database"

Patch: /u02/midia/32226219/28655963
Reason: This patch is not applicable to this specified target type - "oracle_database"

Patch: /u02/midia/32226219/32240606
Reason: This patch is not applicable to this specified target type - "oracle_database"

==Following patches were SUCCESSFULLY applied:

Patch: /u02/midia/32226219/32204699
Log: /u01/app/18/db/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-09-12PM_1.log

Patch: /u02/midia/32226219/32216973
Log: /u01/app/18/db/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-09-12PM_1.log

Host:upgrade
SIHA Home:/u01/app/18/grid
Version:18.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/midia/32226219/28655963
Log: /u01/app/18/grid/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-19-32PM_1.log

Patch: /u02/midia/32226219/32204699
Log: /u01/app/18/grid/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-19-32PM_1.log

Patch: /u02/midia/32226219/32216944
Log: /u01/app/18/grid/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-19-32PM_1.log

Patch: /u02/midia/32226219/32216973
Log: /u01/app/18/grid/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-19-32PM_1.log

Patch: /u02/midia/32226219/32240606
Log: /u01/app/18/grid/cfgtoollogs/patchauto/core/patch/patch2021-03-09_17-19-32PM_1.log

OPatchauto session completed at Tue Mar 9 17:40:55 2021
Time taken to complete the session 34 minutes, 30 seconds

Evidências do PSU aplicado:

```
SQL> select comp_name, version, status, modified from dba_registry;
```

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Database Catalog Views	18.0.0.0.0	VALID	05-MAR-2021 21:56:51
Oracle Database Packages and Types	18.0.0.0.0	VALID	05-MAR-2021 21:56:51
Oracle Real Application Clusters	18.0.0.0.0	OPTION OFF	07-FEB-2018 21:05:23
JServer JAVA Virtual Machine	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle XDK	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Database Java Packages	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
OLAP Analytic Workspace	18.0.0.0.0	VALID	09-MAR-2021 17:39:19
Oracle XML Database	18.0.0.0.0	VALID	05-MAR-2021 21:56:53
Oracle Workspace Manager	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Text	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Oracle Multimedia	18.0.0.0.0	VALID	05-MAR-2021 21:56:54
Spatial	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle OLAP API	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle Label Security	18.0.0.0.0	VALID	05-MAR-2021 21:56:56
Oracle Database Vault	18.0.0.0.0	VALID	05-MAR-2021 21:56:56

15 rows selected.

```
SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;
```

SP2-0734: unknown command beginning "jselect PA..." - rest of line ignored.

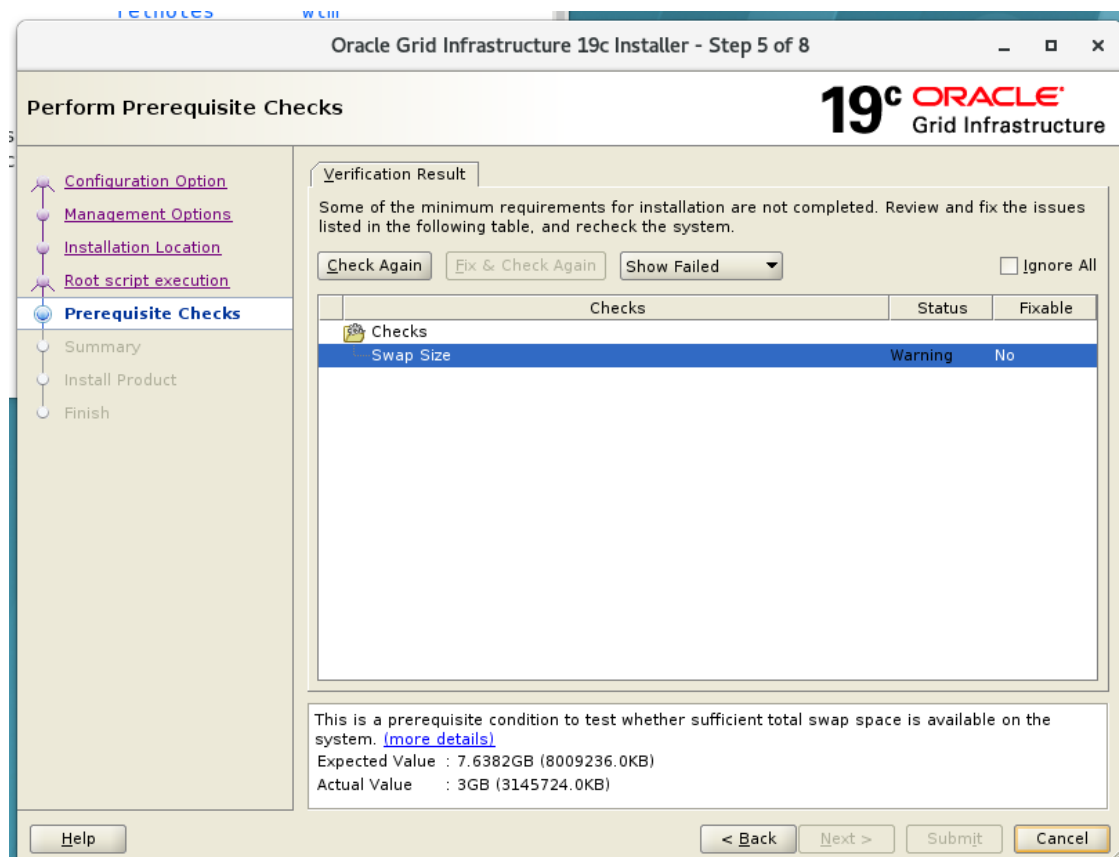
```
SQL>
```

```
SQL>
```

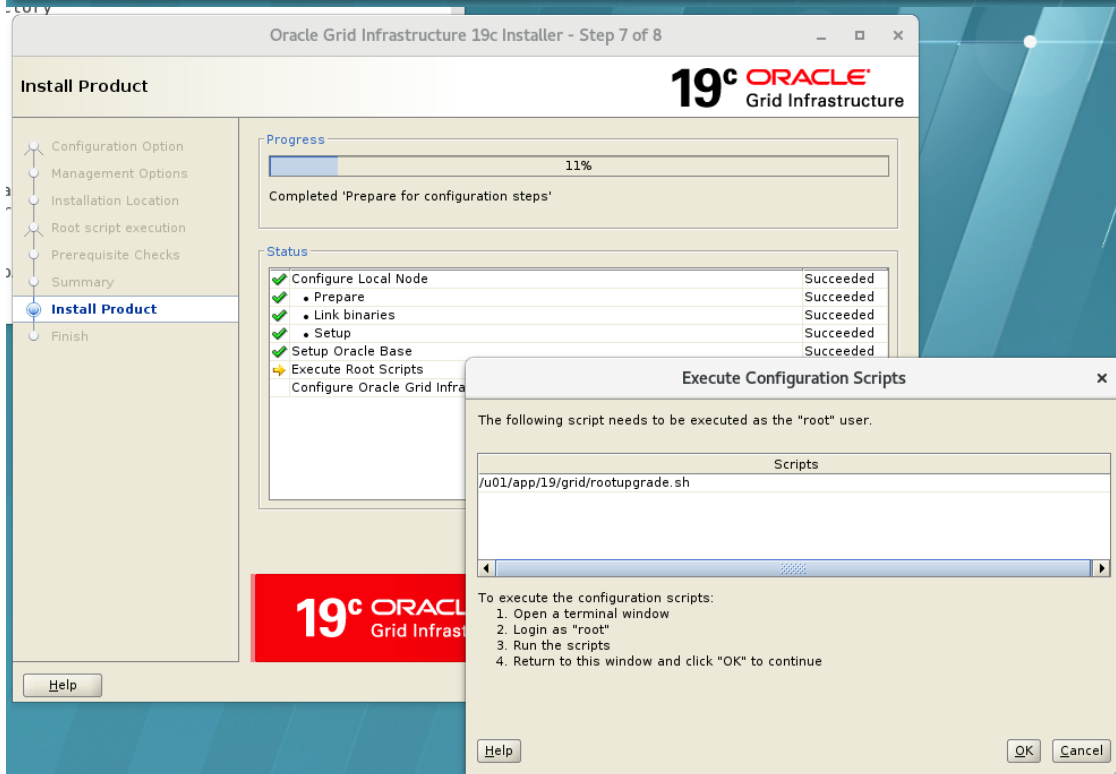
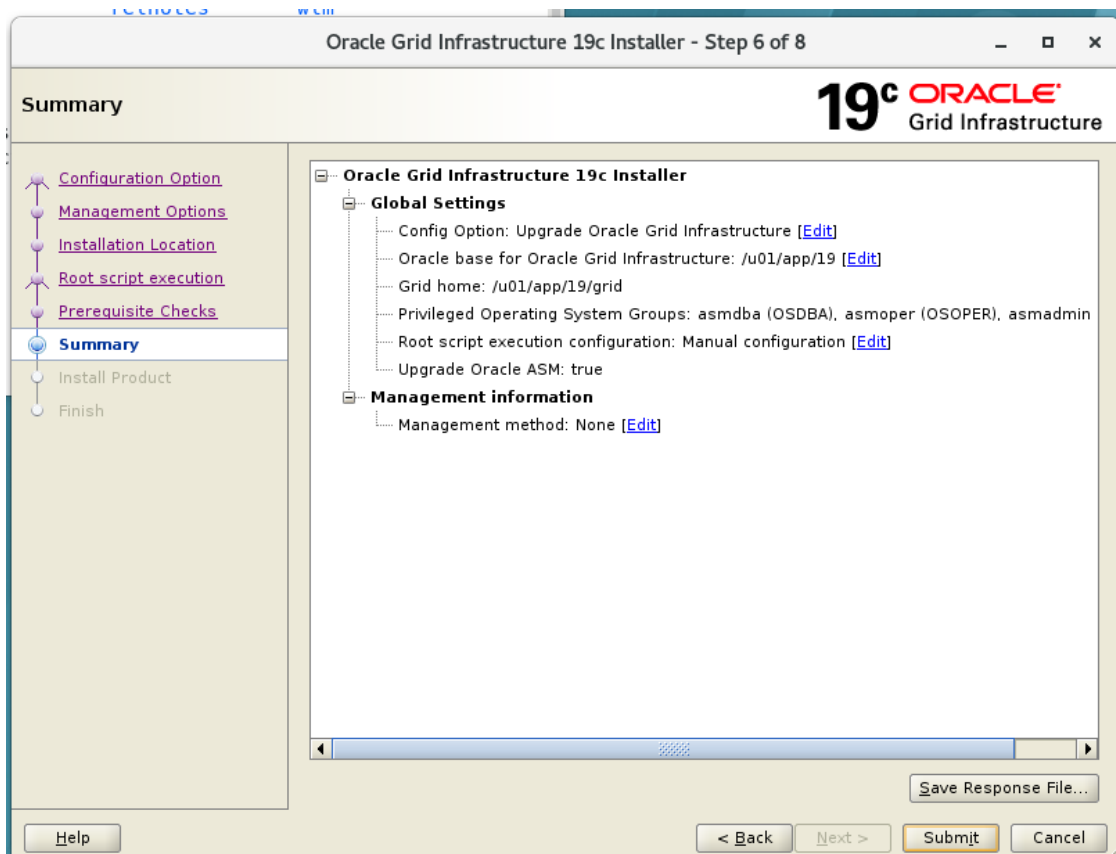
```
SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;
```

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
28090523	APPLY	SUCCESS	05-MAR-21 09:55:24.957775 PM	Database Release Update : 18.3.0.0.180717 (28090523)
27923415	APPLY	SUCCESS	05-MAR-21 09:55:24.963174 PM	OJVM RELEASE UPDATE: 18.3.0.0.180717 (27923415)
32204699	APPLY	SUCCESS	09-MAR-21 05:40:54.146445 PM	Database Release Update : 18.13.0.0.210119 (32204699)

4- Configuração do Grid (Tentativa após PSU).



Clicamos em Ignore All e seguimos em Next>



Primeira tentativa de rodar o rootupgrade.sh deu erro na hora de subir o ohasd.

```
[root@upgrade 19]# /u01/app/19/grid/rootupgrade.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19/grid
```

```
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
```

```
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
```

Now product-specific root actions will be performed.

Using configuration parameter file: /u01/app/19/grid/crs/install/crsconfig_params

The log of current session can be found at:

```
/u01/app/19/crsdata/upgrade/crsconfig/roothas_2021-03-09_06-03-37PM.log
2021/03/09 18:03:38 CLSRSC-595: Executing upgrade step 1 of 12: 'UpgPrechecks'.
2021/03/09 18:03:42 CLSRSC-363: User ignored prerequisites during installation
2021/03/09 18:03:43 CLSRSC-595: Executing upgrade step 2 of 12: 'GetOldConfig'.
CRS-6705: Oracle Clusterware Release Version ('19.0.0.0.0') does not match Software Version ('18.0.0.0.0'). Oracle Clusterware cannot be started
.
```

CRS-4000: Command Start failed, or completed with errors.

```
2021/03/09 18:04:01 CLSRSC-318: Failed to start Oracle OHASD service
Died at /u01/app/19/grid/crs/install/crsupgrade.pm line 6269.
```

Sendo assim, subi ele na mão, ainda com as variáveis do ambiente antigo e, em seguida, rodei o rootupgrade.sh:

```
[root@upgrade grid]# . /home/grid/grid_profile
[root@upgrade grid]# crsctl start has
CRS-4123: Oracle High Availability Services has been started.
```

rootupgrade.sh

```
[root@upgrade grid]# /u01/app/19/grid/rootupgrade.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19/grid
```

```
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
```

```
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Using configuration parameter file: /u01/app/19/grid/crs/install/crsconfig_params
The log of current session can be found at:
```

```
/u01/app/19/crsdata/upgrade/crsconfig/roothas_2021-03-09_06-13-26PM.log
2021/03/09 18:13:28 CLSRSC-595: Executing upgrade step 1 of 12: 'UpgPrechecks'.
2021/03/09 18:13:32 CLSRSC-363: User ignored prerequisites during installation
2021/03/09 18:13:33 CLSRSC-595: Executing upgrade step 2 of 12: 'GetOldConfig'.
2021/03/09 18:13:35 CLSRSC-595: Executing upgrade step 3 of 12: 'GenSiteGUIDs'.
2021/03/09 18:13:35 CLSRSC-595: Executing upgrade step 4 of 12: 'SetupOSD'.
2021/03/09 18:13:35 CLSRSC-595: Executing upgrade step 5 of 12: 'PreUpgrade'.
```

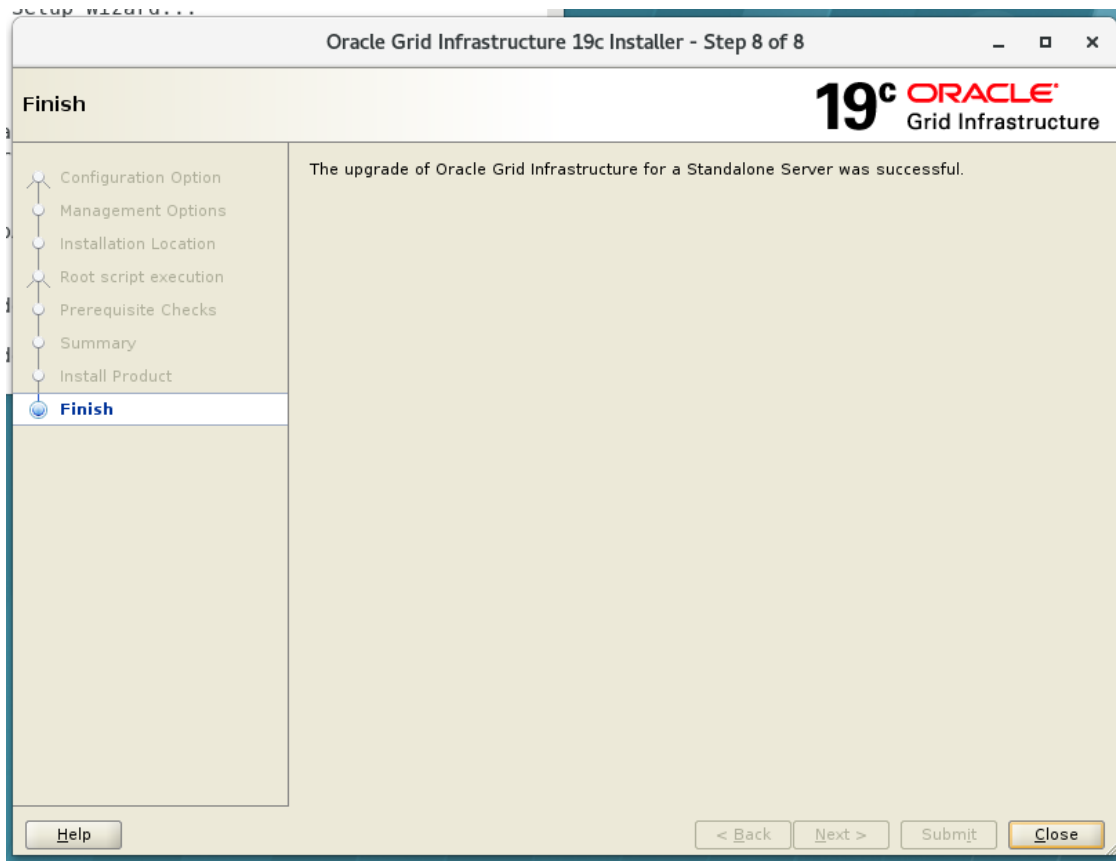
ASM has been upgraded and started successfully.

```
2021/03/09 18:14:42 CLSRSC-595: Executing upgrade step 6 of 12: 'UpgradeAFD'.
2021/03/09 18:14:43 CLSRSC-595: Executing upgrade step 7 of 12: 'UpgradeOLR'.
clscfg: EXISTING configuration version 0 detected.
Creating OCR keys for user 'grid', privgrp 'oinstall'..
Operation successful.
2021/03/09 18:14:49 CLSRSC-595: Executing upgrade step 8 of 12: 'UpgradeOCR'.
LOCAL ONLY MODE
Successfully accumulated necessary OCR keys.
Creating OCR keys for user 'root', privgrp 'root'..
Operation successful.
CRS-4664: Node upgrade successfully pinned.
2021/03/09 18:14:52 CLSRSC-595: Executing upgrade step 9 of 12: 'CreateOHASD'.
2021/03/09 18:14:53 CLSRSC-595: Executing upgrade step 10 of 12: 'ConfigOHASD'.
2021/03/09 18:14:53 CLSRSC-329: Replacing Clusterware entries in file 'oracle-ohasd.service'
2021/03/09 18:15:44 CLSRSC-595: Executing upgrade step 11 of 12: 'UpgradeSIHA'.
```

```
upgrade 2021/03/09 18:16:50 /u01/app/19/crsdata/upgrade/olr/backup_20210309_181650.olr 724960844
```

```
upgrade 2021/03/05 20:48:54 /u01/app/18/grid/cdata/upgrade/backup_20210305_204854.olr 70732493
```

```
2021/03/09 18:16:50 CLSRSC-595: Executing upgrade step 12 of 12: 'InstallACFS'.
2021/03/09 18:20:03 CLSRSC-327: Successfully configured Oracle Restart for a standalone server
```



Instalação da camada de grid concluída com sucesso.
Note que neste ponto a camada do grid já está no ar:

```
[grid@upgrade ~]$ . grid_profile_19c
[grid@upgrade ~]$ crsctl status res -t
```

Name	Target	State	Server	State details	

Local Resources					

ora.ARCH_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.DATA_AUX_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.DATA_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.LISTENER.lsnr	ONLINE	ONLINE	upgrade	STABLE	
ora.asm	ONLINE	ONLINE	upgrade	Started,STABLE	
ora.ons	OFFLINE	OFFLINE	upgrade	STABLE	

Cluster Resources					

ora.cssd	1	ONLINE	ONLINE	upgrade	STABLE
ora.diskmon	1	OFFLINE	OFFLINE		STABLE
ora.evmd	1	ONLINE	ONLINE	upgrade	STABLE
ora.oraex.db	1	OFFLINE	OFFLINE		Instance Shutdown,ST
ABLE					

```
[grid@upgrade ~]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 18.0.0.0.0 - Production on Tue Mar 9 18:30:00 2021
Version 18.13.0.0.0
```

```
Copyright (c) 1982, 2018, Oracle. All rights reserved.
```

```
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL>
```

Ações de ASM:


```

[grid@upgrade ~]$ srvctl status asm
ASM is running on upgrade
[grid@upgrade ~]$ sqlplus / as sysasm

SQL*Plus: Release 18.0.0.0.0 - Production on Tue Mar 9 18:46:49 2021
Version 18.13.0.0.0

Copyright (c) 1982, 2018, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> col VALUE format a30;
col COMPATIBILITY form a10
col DATABASE_COMPATIBILITY form a10
col NAME form a20
select group_number, name, compatibility, database_compatibility from v$asm_diskgroup;

GROUP_NUMBER NAME                COMPATIBIL DATABASE_C
-----
1 ARCH_ORAEX      18.0.0.0.0 10.1.0.0.0
2 DATA_AUX_ORAEX 18.0.0.0.0 10.1.0.0.0
3 DATA_ORAEX      18.0.0.0.0 10.1.0.0.0

SQL> select g.name,a.name,a.value from v$asm_diskgroup g, v$asm_attribute a where g.group_number=a.group_number and a.name like '%com
pat%';

NAME                NAME                VALUE
-----
ARCH_ORAEX          compatible.asm      18.0.0.0.0
ARCH_ORAEX          compatible.rdbms    10.1.0.0.0
ARCH_ORAEX          compatible.advm     18.0.0.0.0
DATA_AUX_ORAEX      compatible.asm      18.0.0.0.0
DATA_AUX_ORAEX      compatible.rdbms    10.1.0.0.0
DATA_AUX_ORAEX      compatible.advm     18.0.0.0.0
DATA_ORAEX          compatible.asm      18.0.0.0.0
DATA_ORAEX          compatible.rdbms    10.1.0.0.0
DATA_ORAEX          compatible.advm     18.0.0.0.0

9 rows selected.

SQL> alter diskgroup ARCH_ORAEX SET attribute 'compatible.asm'='19.0.0.0.0';

Diskgroup altered.

SQL> alter diskgroup ARCH_ORAEX set attribute 'compatible.rdbms'='18.0.0.0.0';

Diskgroup altered.

SQL> alter diskgroup DATA_AUX_ORAEX SET attribute 'compatible.asm'='19.0.0.0.0';

Diskgroup altered.

SQL> alter diskgroup DATA_AUX_ORAEX set attribute 'compatible.rdbms'='18.0.0.0.0';

Diskgroup altered.

SQL> alter diskgroup DATA_ORAEX SET attribute 'compatible.asm'='19.0.0.0.0';

Diskgroup altered.

SQL> alter diskgroup DATA_ORAEX set attribute 'compatible.rdbms'='18.0.0.0.0';

Diskgroup altered.

```

Upgrade da Camada de Banco

```
[root@upgrade OPatch]# su - oracle
Last login: Tue Mar 9 18:51:22 -03 2021 on :2
[oracle@upgrade ~]$ . db_profile_19c
[oracle@upgrade ~]$ cd $ORACLE_HOME
[oracle@upgrade db]$ pwd
/u01/app/19/db
[oracle@upgrade db]$ ./runInstaller
```

Oracle Database 19c Installer - Step 1 of 9

Select Configuration Option

19^c ORACLE[®] Database

Configuration Option

Database Installation Options

Install Type

Typical Installation

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select any of the following install options.

☐ Create and configure a single instance database.
This option creates a starter database.

☒ **Set Up Software Only**

Note 1: For RAC install, do 'Set Up Software Only' and then execute DBCA (Database Configuration Assistant) from the oracle home.
Note 2: To upgrade an Oracle Database, do 'Set Up Software Only' and then execute DBUA (Database Upgrade Assistant) from the oracle home.

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 2 of 9

Select Database Installation Option

19^c ORACLE[®] Database

Configuration Option

Database Installation Options

Install Type

Typical Installation

Root script execution

Prerequisite Checks

Summary

Install Product

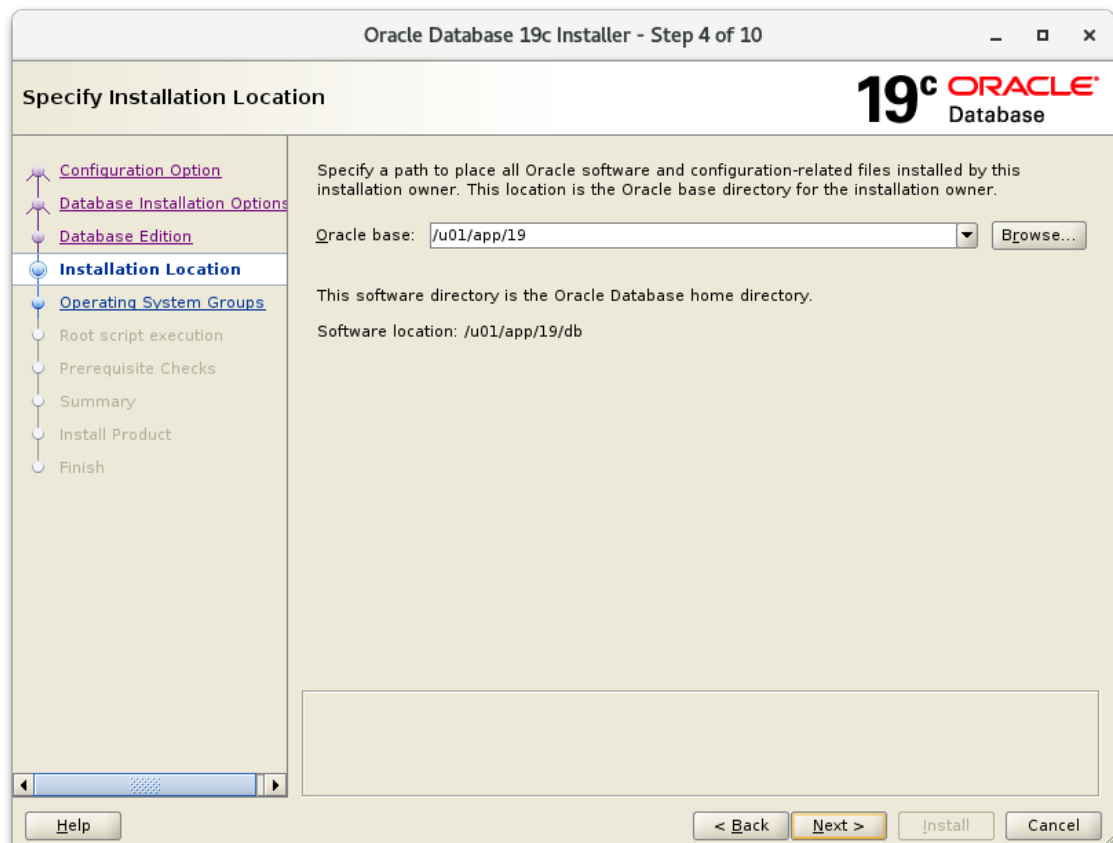
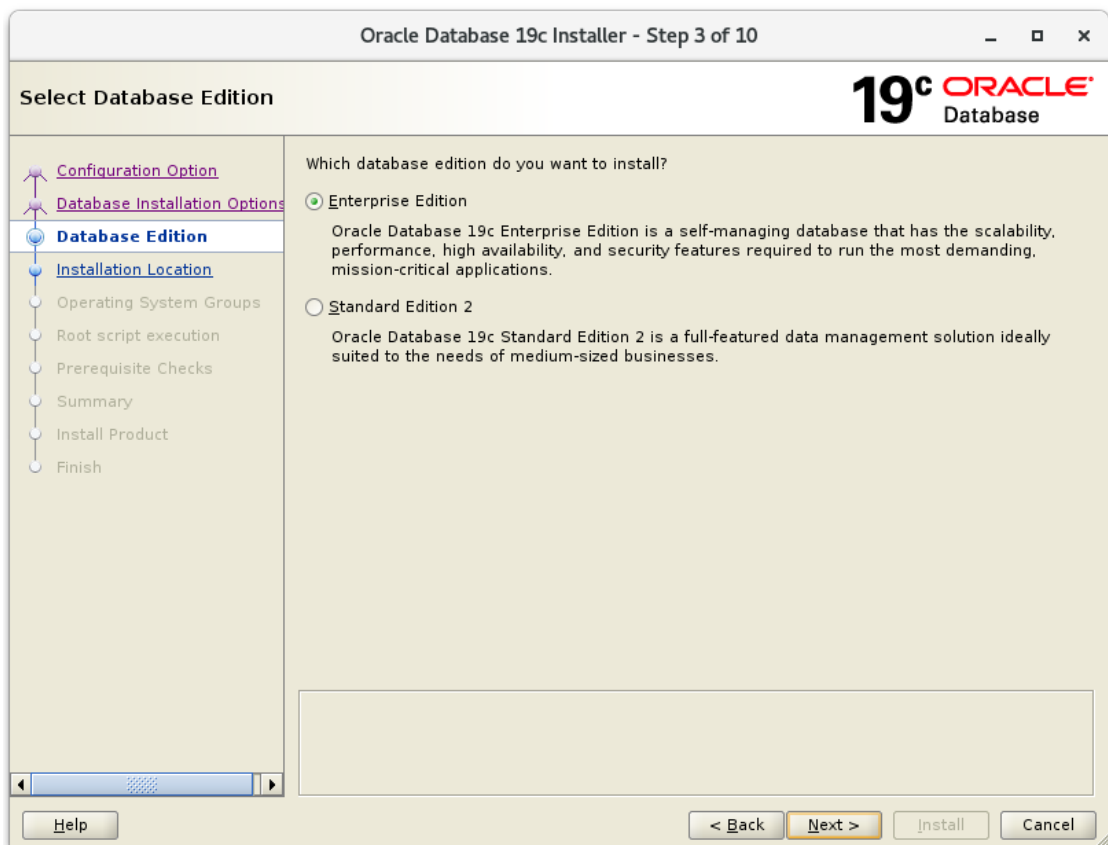
Finish

Select the type of database installation you want to perform.

☒ Single instance database installation

☐ Oracle Real Application Clusters database installation

Help < Back Next > Install Cancel



Oracle Database 19c Installer - Step 5 of 10

19c ORACLE Database

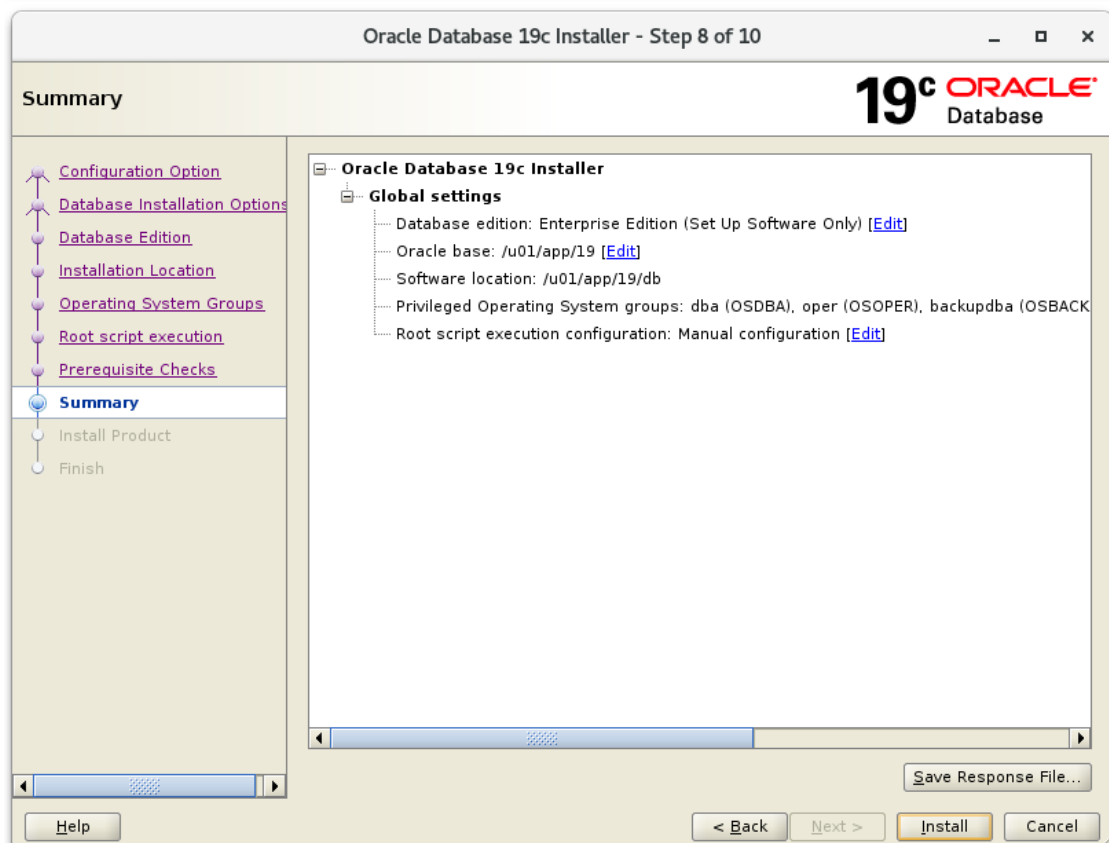
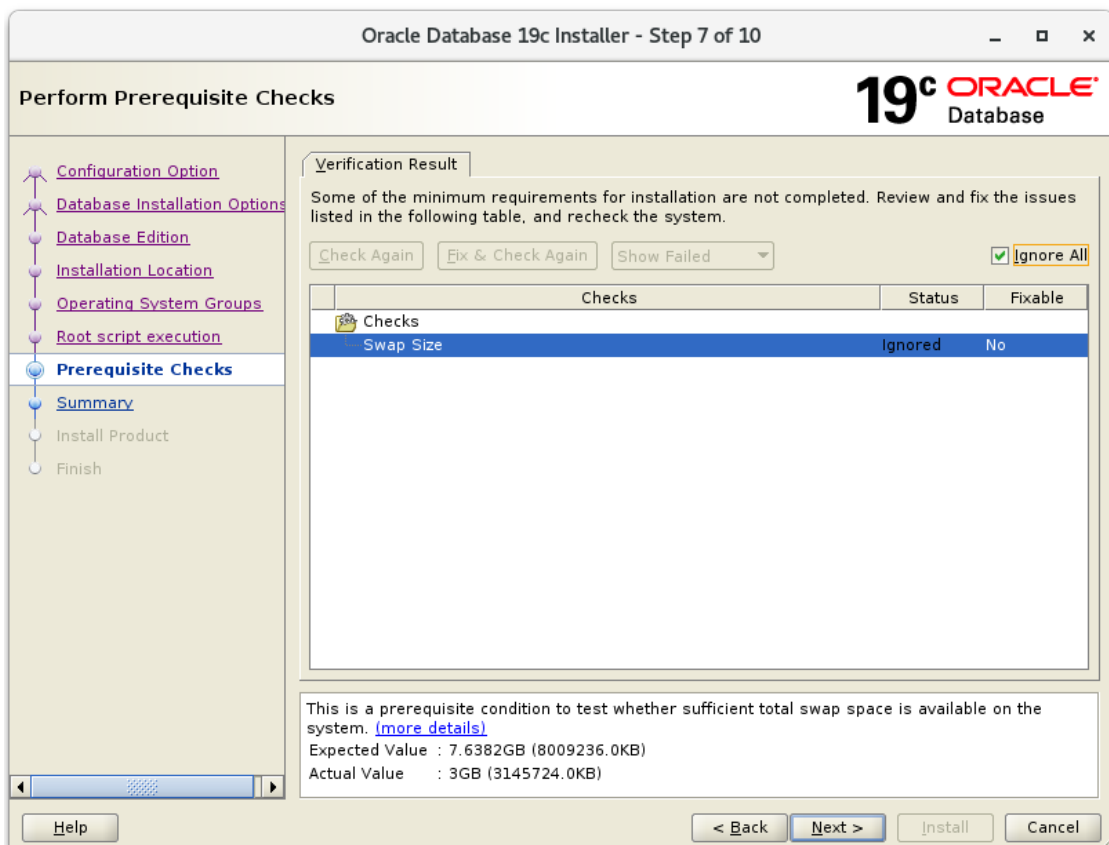
Privileged Operating System groups

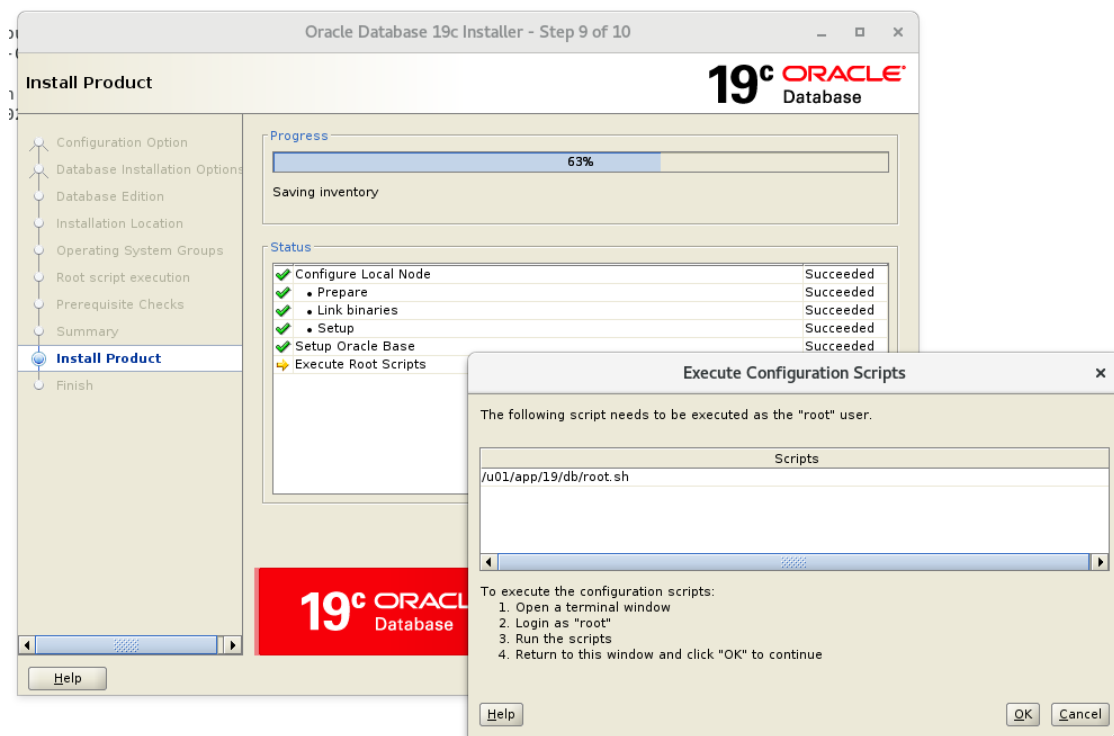
SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.

Database <u>A</u> ddministrator (OSDBA) group:	dba
Database <u>O</u> perator (OSOPER) group (Optional):	oper
Database Backup and Recovery (OSBACKUPDBA) group:	dba
Data <u>G</u> uard administrative (OSDGDBA) group:	dba
Encryption <u>K</u> ey Management administrative (OSKMDBA) group:	dba
<u>R</u> eal Application Cluster administrative (OSRACDBA) group:	dba

Navigation: Configuration Option, Database Installation Options, Database Edition, Installation Location, **Operating System Groups**, Root script execution, Prerequisite Checks, Summary, Install Product, Finish

Buttons: Help, < Back, Next >, Install, Cancel





```
[root@upgrade grid]# /u01/app/19/db/root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/19/db
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:

The contents of "dbhome" have not changed. No need to overwrite.

The contents of "oraenv" have not changed. No need to overwrite.

The contents of "coraenv" have not changed. No need to overwrite.

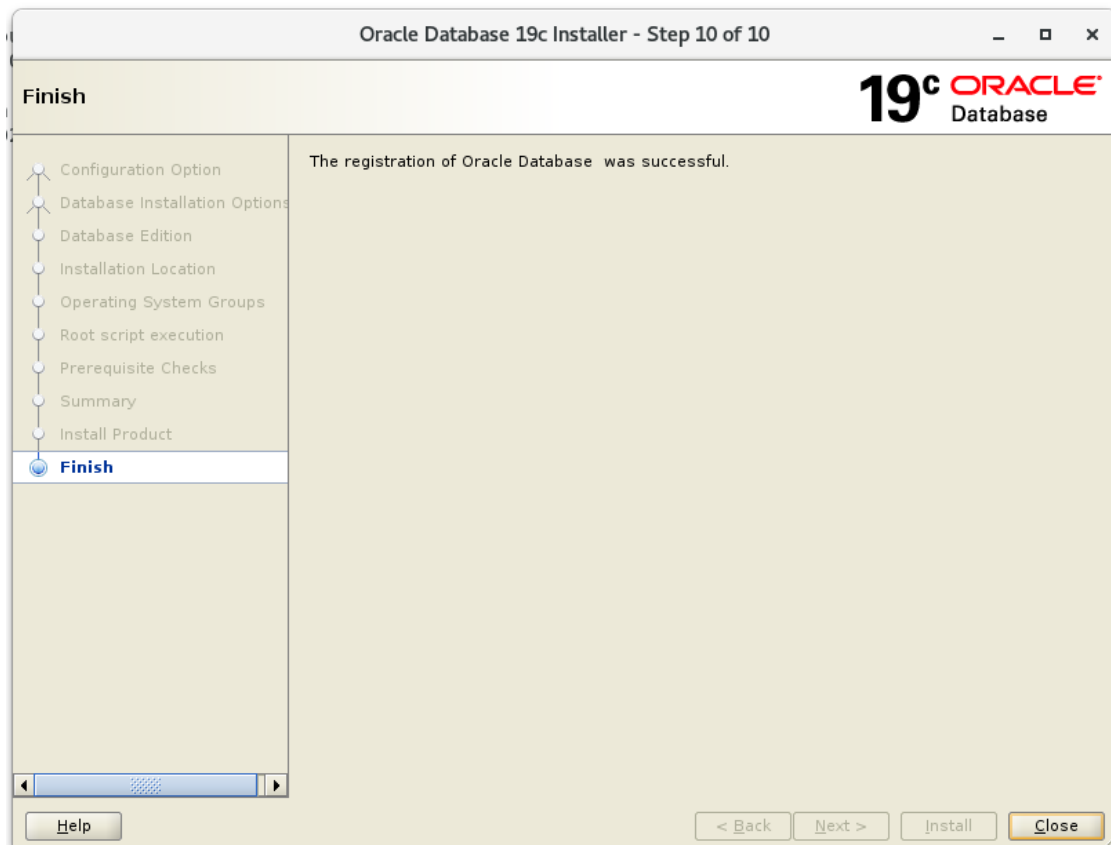
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.

Now product-specific root actions will be performed.

Oracle Trace File Analyzer (TFA - Standalone Mode) is available at :
/u01/app/19/db/bin/tfactl

Note :

1. tfactl will use TFA Service if that service is running and user has been granted access
2. tfactl will configure TFA Standalone Mode only if user has no access to TFA Service or TFA is not installed



Subindo o banco ainda em 18c:

```
[oracle@upgrade ~]$ . db_profile
[oracle@upgrade ~]$ sqlplus / as sysdba

SQL*Plus: Release 18.0.0.0.0 - Production on Tue Mar 9 19:12:29 2021
Version 18.13.0.0.0

Copyright (c) 1982, 2018, Oracle. All rights reserved.

Connected to an idle instance.

SQL> startup
ORACLE instance started.

Total System Global Area 2466248392 bytes
Fixed Size      8660680 bytes
Variable Size   671088640 bytes
Database Buffers 1778384896 bytes
Redo Buffers    8114176 bytes
Database mounted.
Database opened.
SQL> col host_name format a20
SET LINES 200
select UPPER(instance_name) ins_name, host_name, database_status, version, status, to_char(startup_time,'dd-mm-yyyy hh24:mi') Started, login
s
from v$instance
/SQL> SQL> 2 3
```

INS_NAME	HOST_NAME	DATABASE_STATUS	VERSION	STATUS	STARTED	LOGINS
ORAEX	upgrade.localdomain	ACTIVE	18.0.0.0.0	OPEN	09-03-2021 19:12	ALLOWED

Pré upgrade

```
[oracle@upgrade ~]$ . db_profile
[oracle@upgrade ~]$ cd $ORACLE_HOME/jdk/bin
[oracle@upgrade bin]$ java -version
openjdk version "1.8.0_131"
OpenJDK Runtime Environment (build 1.8.0_131-b12)
OpenJDK 64-Bit Server VM (build 25.131-b12, mixed mode)
```

```
[oracle@upgrade bin]$ cd /u01/app/19/db/rdbms/admin/
[oracle@upgrade admin]$ ls -la preupgrade.jar
-rw-r--r--. 1 oracle oinstall 725089 Apr 17 2019 preupgrade.jar
[oracle@upgrade admin]$ chmod 775 preupgrade.jar

[oracle@upgrade admin]$ $ORACLE_HOME/jdk/bin/java -jar /u01/app/19/db/rdbms/admin/preupgrade.jar
Report generated by Oracle Database Pre-Upgrade Information Tool Version
19.0.0.0.0 Build: 1 on 2021-03-09T19:14:54
```

Upgrade-To version: 19.0.0.0.0

=====
Status of the database prior to upgrade
=====

Database Name: ORAEX
Container Name: ORAEX
Container ID: 0
Version: 18.0.0.0.0
DB Patch Level: Database Release Update : 18.13.0.0.210119 (32204699)
Compatible: 18.0.0
Blocksize: 8192
Platform: Linux x86 64-bit
Timezone File: 31
Database log mode: NOARCHIVELOG
Readonly: FALSE
Edition: EE

Oracle Component	Upgrade Action	Current Status
Oracle Server	[to be upgraded]	VALID
JServer JAVA Virtual Machine	[to be upgraded]	VALID
Oracle XDK for Java	[to be upgraded]	VALID
Real Application Clusters	[to be upgraded]	OPTION OFF
Oracle Workspace Manager	[to be upgraded]	VALID
OLAP Analytic Workspace	[to be upgraded]	VALID
Oracle Label Security	[to be upgraded]	VALID
Oracle Database Vault	[to be upgraded]	VALID
Oracle Text	[to be upgraded]	VALID
Oracle XML Database	[to be upgraded]	VALID
Oracle Java Packages	[to be upgraded]	VALID
Oracle Multimedia	[to be upgraded]	VALID
Oracle Spatial	[to be upgraded]	VALID
Oracle OLAP API	[to be upgraded]	VALID

=====
BEFORE UPGRADE
=====

REQUIRED ACTIONS
=====
None

INFORMATION ONLY
=====

- 1. To help you keep track of your tablespace allocations, the following AUTOEXTEND tablespaces are expected to successfully EXTEND during the upgrade process.

Tablespace	Min Size	Size For Upgrade
SYSAUX	590 MB	599 MB
SYSTEM	980 MB	1095 MB
TEMP	85 MB	150 MB
UNDOTBS1	335 MB	439 MB

- Minimum tablespace sizes for upgrade are estimates.
- 2. Check the Oracle Backup and Recovery User's Guide for information on how

to manage an RMAN recovery catalog schema.

If you are using a version of the recovery catalog schema that is older than that required by the RMAN client version, then you must upgrade the catalog schema.

It is good practice to have the catalog schema the same or higher version than the RMAN client version you are using.

=====
AFTER UPGRADE
=====

REQUIRED ACTIONS

=====
None

RECOMMENDED ACTIONS

=====

3. Upgrade the database time zone file using the DBMS_DST package.

The database is using time zone file version 31 and the target 19 release ships with time zone file version 32.

Oracle recommends upgrading to the desired (latest) version of the time zone file. For more information, refer to "Upgrading the Time Zone File and Timestamp with Time Zone Data" in the 19 Oracle Database Globalization Support Guide.

4. To identify directory objects with symbolic links in the path name, run \$ORACLE_HOME/rdbms/admin/utldirsymlink.sql AS SYSDBA after upgrade. Recreate any directory objects listed, using path names that contain no symbolic links.

Some directory object path names may currently contain symbolic links.

Starting in Release 18c, symbolic links are not allowed in directory object path names used with BFILE data types, the UTL_FILE package, or external tables.

5. (AUTOFIXUP) Gather dictionary statistics after the upgrade using the command:

```
EXECUTE DBMS_STATS.GATHER_DICTIONARY_STATS;
```

Oracle recommends gathering dictionary statistics after upgrade.

Dictionary statistics provide essential information to the Oracle optimizer to help it find efficient SQL execution plans. After a database upgrade, statistics need to be re-gathered as there can now be tables that have significantly changed during the upgrade or new tables that do not have statistics gathered yet.

6. Gather statistics on fixed objects after the upgrade and when there is a representative workload on the system using the command:

```
EXECUTE DBMS_STATS.GATHER_FIXED_OBJECTS_STATS;
```

This recommendation is given for all preupgrade runs.

Fixed object statistics provide essential information to the Oracle optimizer to help it find efficient SQL execution plans. Those statistics are specific to the Oracle Database release that generates them, and can be stale upon database upgrade.

For information on managing optimizer statistics, refer to the 18.0.0.0 Oracle Database Upgrade Guide.

ORACLE GENERATED FIXUP SCRIPT
=====

All of the issues in database ORAEX
which are identified above as AFTER UPGRADE "(AUTOFIXUP)" can be resolved by
executing the following

```
SQL>@/u01/app/18/cfgtoollogs/ORAEX/preupgrade/postupgrade_fixups.sql
```

```
=====
PREUPGRADE SUMMARY
=====
```

```
/u01/app/18/cfgtoollogs/ORAEX/preupgrade/preupgrade.log
/u01/app/18/cfgtoollogs/ORAEX/preupgrade/preupgrade_fixups.sql
/u01/app/18/cfgtoollogs/ORAEX/preupgrade/postupgrade_fixups.sql
```

Execute fixup scripts as indicated below:

Before upgrade:

Log into the database and execute the preupgrade fixups
@/u01/app/18/cfgtoollogs/ORAEX/preupgrade/preupgrade_fixups.sql

After the upgrade:

Log into the database and execute the postupgrade fixups
@/u01/app/18/cfgtoollogs/ORAEX/preupgrade/postupgrade_fixups.sql

Preupgrade complete: 2021-03-09T19:14:54
[oracle@upgrade admin]\$

Executando o script de pré upgrade: @/u01/app/18/cfgtoollogs/ORAEX/preupgrade/preupgrade_fixups.sql

```
[oracle@upgrade admin]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 18.0.0.0.0 - Production on Wed Mar 10 11:08:39 2021
Version 18.13.0.0.0
```

```
Copyright (c) 1982, 2018, Oracle. All rights reserved.
```

```
Connected to:
```

```
Oracle Database 18c Enterprise Edition Release 18.0.0.0.0 - Production
Version 18.13.0.0.0
```

```
SQL> @/u01/app/18/cfgtoollogs/ORAEX/preupgrade/preupgrade_fixups.sql
Executing Oracle PRE-Upgrade Fixup Script
```

```
Auto-Generated by:   Oracle Preupgrade Script
                    Version: 19.0.0.0.0 Build: 1
Generated on:        2021-03-10 11:08:08
```

```
For Source Database:   ORAEX
Source Database Version: 18.0.0.0.0
For Upgrade to Version: 19.0.0.0.0
```

Preup Action Number	Preupgrade Check Name	Preupgrade Issue Is	Remedied	Further DBA Action
1.	tablespaces_info	NO	Informational only. Further action is optional.	
2.	rman_recovery_version	NO	Informational only. Further action is optional.	

The fixup scripts have been run and resolved what they can. However, there are still issues originally identified by the preupgrade that have not been remedied and are still present in the database. Depending on the severity of the specific issue, and the nature of the issue itself, that could mean that your database is not ready for upgrade. To resolve the outstanding issues, start by reviewing the preupgrade_fixups.sql and searching it for the name of the failed CHECK NAME or Preupgrade Action Number listed above. There you will find the original corresponding diagnostic message from the preupgrade which explains in more detail what still needs to be done.

```
PL/SQL procedure successfully completed.
```

Para não ter erro, criar um pfile novo, mais recente e baixar o banco:

```
SQL> create pfile='/u02/midia/initDB_18c.ora' from spfile;
```

```
File created.
```

```
SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
```

Subindo o banco em 19c:

```
[oracle@upgrade ~]$ . db_profile_19c
[oracle@upgrade ~]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Mar 10 11:11:21 2021
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

Connected to an idle instance.

```
SQL> create spfile from pfile='/u02/midia/initDB_18c.ora';
```

File created.

```
SQL> startup upgrade;  
ORACLE instance started.
```

Total System Global Area 2466249672 bytes

Fixed Size 8899528 bytes

Variable Size 587202560 bytes

Database Buffers 1862270976 bytes

Redo Buffers 7876608 bytes

Database mounted.

Database opened.

```
SQL> exit
```

Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Version 19.3.0.0.0

```
[oracle@upgrade ~]$ cd $ORACLE_HOME/rdbms/admin
```

```
[oracle@upgrade admin]$ $ORACLE_HOME/perl/bin/perl catctl.pl -n 8 catupgrd.sql
```

Argument list for [catctl.pl]

For Oracle internal use only A = 0

Run in c = 0

Do not run in C = 0

Input Directory d = 0

Echo OFF e = 1

Simulate E = 0

Forced cleanup F = 0

Log Id i = 0

Child Process l = 0

Log Dir l = 0

Priority List Name L = 0

Upgrade Mode active M = 0

SQL Process Count n = 8

SQL PDB Process Count N = 0

Open Mode Normal o = 0

Start Phase p = 0

End Phase P = 0

Reverse Order r = 0

AutoUpgrade Resume R = 0

Script s = 0

Serial Run S = 0

RO User Tablespaces T = 0

Display Phases y = 0

Debug catcon.pm z = 0

Debug catctl.pl Z = 0

catctl.pl VERSION: [19.0.0.0.0]

STATUS: [Production]

BUILD: [RDBMS_19.3.0.0.0DBRU_LINUX.X64_190417]

/u01/app/19/db/rdbms/admin/orahome = [/u01/app/19/db]

/u01/app/19/db/bin/orabasehome = [/u01/app/19/db]

catctlGetOraBaseLogDir = [/u01/app/19/db]

Analyzing file /u01/app/19/db/rdbms/admin/catupgrd.sql

Log file directory = [/tmp/cfgtoollogs/upgrade20210309192741]

catcon::set_log_file_base_path: ALL catcon-related output will be written to [/tmp/cfgtoollogs/upgrade20210309192741/catupgrd_catcon_22465.lst]

catcon::set_log_file_base_path: catcon: See [/tmp/cfgtoollogs/upgrade20210309192741/catupgrd*.log] files for output generated by scripts

catcon::set_log_file_base_path: catcon: See [/tmp/cfgtoollogs/upgrade20210309192741/catupgrd_*.lst] files for spool files, if any

Number of CPUs = 2

```

Database Name      = ORAEX
DataBase Version   = 18.0.0.0.0
catcon::set_log_file_base_path: ALL catcon-related output will be written to [/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753/catupgrd
_catcon_22465.lst]

catcon::set_log_file_base_path: catcon: See [/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753/catupgrd*.log] files for output generate
d by scripts

catcon::set_log_file_base_path: catcon: See [/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753/catupgrd_*.lst] files for spool files, if an
y

Log file directory = [/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753]

Parallel SQL Process Count      = 8
Components in [ORAEX]
  Installed [APS CATALOG CATJAVA CATPROC CONTEXT DV JAVAVM OLS ORDIM OWM SDO XDB XML XOQ]
  Not Installed [APEX EM MGW ODM RAC WK]

-----
Phases [0-107]      Start Time:[2021_03_09 19:28:04]
-----
***** Executing Change Scripts *****
Serial Phase #0 [ORAEX] Files:1 Time: 31s
***** Catalog Core SQL *****
Serial Phase #1 [ORAEX] Files:5 Time: 33s
Restart Phase #2 [ORAEX] Files:1 Time: 3s
***** Catalog Tables and Views *****
Parallel Phase #3 [ORAEX] Files:19 Time: 14s
Restart Phase #4 [ORAEX] Files:1 Time: 2s
***** Catalog Final Scripts *****
Serial Phase #5 [ORAEX] Files:7 Time: 12s
***** Catproc Start *****
Serial Phase #6 [ORAEX] Files:1 Time: 13s
***** Catproc Types *****
Serial Phase #7 [ORAEX] Files:2 Time: 13s
Restart Phase #8 [ORAEX] Files:1 Time: 2s
***** Catproc Tables *****
Parallel Phase #9 [ORAEX] Files:67 Time: 21s
Restart Phase #10 [ORAEX] Files:1 Time: 4s
***** Catproc Package Specs *****
Serial Phase #11 [ORAEX] Files:1 Time: 55s
Restart Phase #12 [ORAEX] Files:1 Time: 2s
***** Catproc Procedures *****
Parallel Phase #13 [ORAEX] Files:94 Time: 13s
Restart Phase #14 [ORAEX] Files:1 Time: 3s
Parallel Phase #15 [ORAEX] Files:120 Time: 18s
Restart Phase #16 [ORAEX] Files:1 Time: 4s
Serial Phase #17 [ORAEX] Files:22 Time: 6s
Restart Phase #18 [ORAEX] Files:1 Time: 3s
***** Catproc Views *****
Parallel Phase #19 [ORAEX] Files:32 Time: 17s
Restart Phase #20 [ORAEX] Files:1 Time: 3s
Serial Phase #21 [ORAEX] Files:3 Time: 11s
Restart Phase #22 [ORAEX] Files:1 Time: 2s
Parallel Phase #23 [ORAEX] Files:25 Time: 171s
Restart Phase #24 [ORAEX] Files:1 Time: 3s
Parallel Phase #25 [ORAEX] Files:12 Time: 121s
Restart Phase #26 [ORAEX] Files:1 Time: 4s
Serial Phase #27 [ORAEX] Files:1 Time: 0s
Serial Phase #28 [ORAEX] Files:3 Time: 4s
Serial Phase #29 [ORAEX] Files:1 Time: 0s
Restart Phase #30 [ORAEX] Files:1 Time: 3s
***** Catproc CDB Views *****
Serial Phase #31 [ORAEX] Files:1 Time: 2s
Restart Phase #32 [ORAEX] Files:1 Time: 4s
Serial Phase #34 [ORAEX] Files:1 Time: 0s
***** Catproc PLBs *****
Serial Phase #35 [ORAEX] Files:293 Time: 28s
Serial Phase #36 [ORAEX] Files:1 Time: 0s

```

```

Restart Phase #:37 [ORAEX] Files:1 Time: 4s
Serial Phase #:38 [ORAEX] Files:6 Time: 7s
Restart Phase #:39 [ORAEX] Files:1 Time: 2s
***** Catproc DataPump *****
Serial Phase #:40 [ORAEX] Files:3 Time: 47s
Restart Phase #:41 [ORAEX] Files:1 Time: 5s
***** Catproc SQL *****
Parallel Phase #:42 [ORAEX] Files:13 Time: 118s
Restart Phase #:43 [ORAEX] Files:1 Time: 2s
Parallel Phase #:44 [ORAEX] Files:11 Time: 8s
Restart Phase #:45 [ORAEX] Files:1 Time: 4s
Parallel Phase #:46 [ORAEX] Files:3 Time: 3s
Restart Phase #:47 [ORAEX] Files:1 Time: 4s
***** Final Catproc scripts *****
Serial Phase #:48 [ORAEX] Files:1 Time: 10s
Restart Phase #:49 [ORAEX] Files:1 Time: 4s
***** Final RDBMS scripts *****
Serial Phase #:50 [ORAEX] Files:1 Time: 4s
***** Upgrade Component Start *****
Serial Phase #:51 [ORAEX] Files:1 Time: 2s
Restart Phase #:52 [ORAEX] Files:1 Time: 3s
***** Upgrading Java and non-Java *****
Serial Phase #:53 [ORAEX] Files:2 Time: 271s
***** Upgrading XDB *****
Restart Phase #:54 [ORAEX] Files:1 Time: 2s
Serial Phase #:56 [ORAEX] Files:3 Time: 10s
Serial Phase #:57 [ORAEX] Files:3 Time: 6s
Parallel Phase #:58 [ORAEX] Files:10 Time: 6s
Parallel Phase #:59 [ORAEX] Files:25 Time: 7s
Serial Phase #:60 [ORAEX] Files:4 Time: 9s
Serial Phase #:61 [ORAEX] Files:1 Time: 0s
Serial Phase #:62 [ORAEX] Files:32 Time: 6s
Serial Phase #:63 [ORAEX] Files:1 Time: 0s
Parallel Phase #:64 [ORAEX] Files:6 Time: 10s
Serial Phase #:65 [ORAEX] Files:2 Time: 20s
Serial Phase #:66 [ORAEX] Files:3 Time: 27s
***** Upgrading ORDIM *****
Restart Phase #:67 [ORAEX] Files:1 Time: 4s
Serial Phase #:69 [ORAEX] Files:1 Time: 5s
Parallel Phase #:70 [ORAEX] Files:2 Time: 29s
Restart Phase #:71 [ORAEX] Files:1 Time: 2s
Parallel Phase #:72 [ORAEX] Files:2 Time: 4s
Serial Phase #:73 [ORAEX] Files:2 Time: 5s
***** Upgrading SDO *****
Restart Phase #:74 [ORAEX] Files:1 Time: 2s
Serial Phase #:76 [ORAEX] Files:1 Time: 40s
Serial Phase #:77 [ORAEX] Files:2 Time: 5s
Restart Phase #:78 [ORAEX] Files:1 Time: 3s
Serial Phase #:79 [ORAEX] Files:1 Time: 46s
Restart Phase #:80 [ORAEX] Files:1 Time: 2s
Parallel Phase #:81 [ORAEX] Files:3 Time: 50s
Restart Phase #:82 [ORAEX] Files:1 Time: 6s
Serial Phase #:83 [ORAEX] Files:1 Time: 8s
Restart Phase #:84 [ORAEX] Files:1 Time: 4s
Serial Phase #:85 [ORAEX] Files:1 Time: 13s
Restart Phase #:86 [ORAEX] Files:1 Time: 4s
Parallel Phase #:87 [ORAEX] Files:4 Time: 74s
Restart Phase #:88 [ORAEX] Files:1 Time: 5s
Serial Phase #:89 [ORAEX] Files:1 Time: 4s
Restart Phase #:90 [ORAEX] Files:1 Time: 3s
Serial Phase #:91 [ORAEX] Files:2 Time: 11s
Restart Phase #:92 [ORAEX] Files:1 Time: 2s
Serial Phase #:93 [ORAEX] Files:1 Time: 2s
Restart Phase #:94 [ORAEX] Files:1 Time: 4s
***** Upgrading ODM, WK, EXF, RUL, XOQ *****
Serial Phase #:95 [ORAEX] Files:1 Time: 14s
Restart Phase #:96 [ORAEX] Files:1 Time: 2s
***** Final Component scripts *****
Serial Phase #:97 [ORAEX] Files:1 Time: 3s
***** Final Upgrade scripts *****

```

```

Serial Phase #:98 [ORAEX] Files:1 Time: 198s
***** Migration *****
Serial Phase #:99 [ORAEX] Files:1 Time: 2s
*** End PDB Application Upgrade Pre-Shutdown ***
Serial Phase #:100 [ORAEX] Files:1 Time: 2s
Serial Phase #:101 [ORAEX] Files:1 Time: 0s
Serial Phase #:102 [ORAEX] Files:1 Time: 102s
***** Post Upgrade *****
Serial Phase #:103 [ORAEX] Files:1 Time: 30s
***** Summary report *****
Serial Phase #:104 [ORAEX] Files:1 Time: 4s
*** End PDB Application Upgrade Post-Shutdown **
Serial Phase #:105 [ORAEX] Files:1 Time: 2s
Serial Phase #:106 [ORAEX] Files:1 Time: 0s
Serial Phase #:107 [ORAEX] Files:1 Time: 41s

```

```

-----
Phases [0-107] End Time:[2021_03_09 20:00:47]
-----

```

Grand Total Time: 1965s

LOG FILES: (/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753/catupgrd*.log)

Upgrade Summary Report Located in:
/u01/app/19/db/cfgtoollogs/ORAEX/upgrade20210309192753/upg_summary.log

Grand Total Upgrade Time: [0d:0h:32m:45s]
[oracle@upgrade admin]\$

Subindo o Banco pós catupgrd.sql

```
[oracle@upgrade admin]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Mar 10 11:44:23 2021
Version 19.3.0.0.0
```

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to an idle instance.

```
SQL> startup;
ORACLE instance started.
```

```
Total System Global Area 2466249672 bytes
Fixed Size 8899528 bytes
Variable Size 654311424 bytes
Database Buffers 1795162112 bytes
Redo Buffers 7876608 bytes
```

Database mounted.

Database opened.

```
SQL> col host_name format a20
```

```
SET LINES 200
```

```
select UPPER(instance_name) ins_name, host_name, database_status, version, status, to_char(startup_time,'dd-mm-yyyy hh24:mi') Started, login
s
```

```
from v$instance
```

```
/SQL> SQL> 2 3
```

INS_NAME	HOST_NAME	DATABASE_STATUS	VERSION	STATUS	STARTED	LOGINS
ORAEX	upgrade.localdomain	ACTIVE	19.0.0.0.0	OPEN	10-03-2021 11:44	ALLOWED

1º Script de Post Upgrade – postupgrade_fixups.sql

```
SQL> @/u01/app/18/cfgtoollogs/ORAEX/preupgrade/postupgrade_fixups.sql
```

Session altered.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

Package created.

No errors.

Package body created.

PL/SQL procedure successfully completed.

No errors.

Package created.

No errors.

Package body created.

No errors.

Executing Oracle POST-Upgrade Fixup Script

Auto-Generated by: Oracle Preupgrade Script

Version: 19.0.0.0.0 Build: 1

Generated on: 2021-03-09 19:14:54

For Source Database: ORAEX

Source Database Version: 18.0.0.0.0

For Upgrade to Version: 19.0.0.0.0

Preup Action Number	Preupgrade Issue Is Preupgrade Check Name	Remedied	Further DBA Action
3.	old_time_zones_exist	NO	Manual fixup recommended.
4.	dir_symlinks	YES	None.
5.	post_dictionary	YES	None.
6.	post_fixed_objects	NO	Informational only. Further action is optional.

The fixup scripts have been run and resolved what they can. However, there are still issues originally identified by the preupgrade that have not been remedied and are still present in the database.

Depending on the severity of the specific issue, and the nature of the issue itself, that could mean that your database upgrade is not fully complete. To resolve the outstanding issues, start by reviewing the postupgrade_fixups.sql and searching it for the name of the failed CHECK NAME or Preupgrade Action Number listed above. There you will find the original corresponding diagnostic message from the preupgrade which explains in more detail what still needs to be done.

PL/SQL procedure successfully completed.

Session altered.

```
SQL>
```

2^o Script de Post Upgrade – utldirsymlink.sql


```
SQL> @$ORACLE_HOME/rdbms/admin/utldirsymlink.sql
No DIRECTORY OBJECTS with symlinks found.
```

PL/SQL procedure successfully completed.

No errors.

3^e Script de Post Upgrade – Stats

```
SQL> EXECUTE DBMS_STATS.GATHER_FIXED_OBJECTS_STATS;
```

PL/SQL procedure successfully completed.

4^e Script de Post Upgrade – utltz_countstar.sql

```
SQL> @?/rdbms/admin/utltz_countstar.sql
```

Session altered.

.

Estimating amount of TSTZ data using COUNT(*).

This might take some time ...

.

For SYS tables first ...

Note: empty tables are not listed.

Owner.TableName.ColumnName - COUNT(*) of that column

```
SYS.AQ$_ALERT_QT_S.CREATION_TIME - 4
SYS.AQ$_ALERT_QT_S.DELETION_TIME - 4
SYS.AQ$_ALERT_QT_S.MODIFICATION_TIME - 4
SYS.AQ$_AQ$_MEM_MC_S.CREATION_TIME - 3
SYS.AQ$_AQ$_MEM_MC_S.DELETION_TIME - 3
SYS.AQ$_AQ$_MEM_MC_S.MODIFICATION_TIME - 3
SYS.AQ$_AQ_PROP_TABLE_S.CREATION_TIME - 1
SYS.AQ$_AQ_PROP_TABLE_S.DELETION_TIME - 1
SYS.AQ$_AQ_PROP_TABLE_S.MODIFICATION_TIME - 1
SYS.AQ$_ORA$PREPLUGIN_BACKUP_QTB_S.CREATION_TIME - 1
SYS.AQ$_ORA$PREPLUGIN_BACKUP_QTB_S.DELETION_TIME - 1
SYS.AQ$_ORA$PREPLUGIN_BACKUP_QTB_S.MODIFICATION_TIME - 1
SYS.AQ$_PDB_MON_EVENT_QTABLE$_S.CREATION_TIME - 1
SYS.AQ$_PDB_MON_EVENT_QTABLE$_S.DELETION_TIME - 1
SYS.AQ$_PDB_MON_EVENT_QTABLE$_S.MODIFICATION_TIME - 1
SYS.AQ$_SCHEDULER$_EVENT_QTAB_S.CREATION_TIME - 3
SYS.AQ$_SCHEDULER$_EVENT_QTAB_S.DELETION_TIME - 3
SYS.AQ$_SCHEDULER$_EVENT_QTAB_S.MODIFICATION_TIME - 3
SYS.AQ$_SCHEDULER$_REMDB_JOBQTAB_S.CREATION_TIME - 1
SYS.AQ$_SCHEDULER$_REMDB_JOBQTAB_S.DELETION_TIME - 1
SYS.AQ$_SCHEDULER$_REMDB_JOBQTAB_S.MODIFICATION_TIME - 1
SYS.AQ$_SCHEDULER_FILEWATCHER_QT_S.CREATION_TIME - 1
SYS.AQ$_SCHEDULER_FILEWATCHER_QT_S.DELETION_TIME - 1
SYS.AQ$_SCHEDULER_FILEWATCHER_QT_S.MODIFICATION_TIME - 1
SYS.AQ$_SUBSCRIBER_TABLE.CREATION_TIME - 1
SYS.AQ$_SUBSCRIBER_TABLE.DELETION_TIME - 1
SYS.AQ$_SUBSCRIBER_TABLE.MODIFICATION_TIME - 1
SYS.AQ$_SYS$SERVICE_METRICS_TAB_S.CREATION_TIME - 4
SYS.AQ$_SYS$SERVICE_METRICS_TAB_S.DELETION_TIME - 4
SYS.AQ$_SYS$SERVICE_METRICS_TAB_S.MODIFICATION_TIME - 4
SYS.KET$_AUTOTASK_STATUS.ABA_START_TIME - 1
SYS.KET$_AUTOTASK_STATUS.ABA_STATE_TIME - 1
SYS.KET$_AUTOTASK_STATUS.MW_RECORD_TIME - 1
SYS.KET$_AUTOTASK_STATUS.MW_START_TIME - 1
SYS.KET$_AUTOTASK_STATUS.RECONCILE_TIME - 1
SYS.KET$_CLIENT_CONFIG.FIELD_2 - 7
SYS.KET$_CLIENT_CONFIG.LAST_CHANGE - 7
SYS.KET$_CLIENT_TASKS.CURR_WIN_START - 3
SYS.KET$_CLIENT_TASKS.LG_DATE - 3
SYS.KET$_CLIENT_TASKS.LT_DATE - 3
SYS.OPTSTAT_HIST_CONTROL$.SPARE6 - 45
SYS.OPTSTAT_HIST_CONTROL$.SVAL2 - 45
SYS.OPTSTAT_SNAPSHOT$.TIMESTAMP - 5543
SYS.OPTSTAT_USER_PREFS$.CHGTIME - 72
SYS.RADM_FPTM$.TSWTZ COL - 1
```

```

SYS.REG$.NTFN_GROUPING_START_TIME - 2
SYS.REG$.REG_TIME - 2
SYS.SCHEDULER$_EVENT_LOG.LOG_DATE - 281
SYS.SCHEDULER$_GLOBAL_ATTRIBUTE.ATTR_TSTAMP - 11
SYS.SCHEDULER$_JOB.END_DATE - 22
SYS.SCHEDULER$_JOB.LAST_ENABLED_TIME - 22
SYS.SCHEDULER$_JOB.LAST_END_DATE - 22
SYS.SCHEDULER$_JOB.LAST_START_DATE - 22
SYS.SCHEDULER$_JOB.NEXT_RUN_DATE - 22
SYS.SCHEDULER$_JOB.START_DATE - 22
SYS.SCHEDULER$_JOB_RUN_DETAILS.LOG_DATE - 60
SYS.SCHEDULER$_JOB_RUN_DETAILS.REQ_START_DATE - 60
SYS.SCHEDULER$_JOB_RUN_DETAILS.START_DATE - 60
SYS.SCHEDULER$_SCHEDULE.END_DATE - 4
SYS.SCHEDULER$_SCHEDULE.REFERENCE_DATE - 4
SYS.SCHEDULER$_WINDOW.ACTUAL_START_DATE - 9
SYS.SCHEDULER$_WINDOW.END_DATE - 9
SYS.SCHEDULER$_WINDOW.LAST_START_DATE - 9
SYS.SCHEDULER$_WINDOW.MANUAL_OPEN_TIME - 9
SYS.SCHEDULER$_WINDOW.NEXT_START_DATE - 9
SYS.SCHEDULER$_WINDOW.START_DATE - 9
SYS.SCHEDULER$_WINDOW_DETAILS.LOG_DATE - 2
SYS.SCHEDULER$_WINDOW_DETAILS.REQ_START_DATE - 2
SYS.SCHEDULER$_WINDOW_DETAILS.START_DATE - 2
SYS.STATS_TARGET$.END_TIME - 986
SYS.STATS_TARGET$.START_TIME - 986
SYS.TAB_STATS$.SPARE6 - 1181
SYS.WRI$_ALERT_HISTORY.CREATION_TIME - 15
SYS.WRI$_ALERT_HISTORY.TIME_SUGGESTED - 15
SYS.WRI$_OPTSTAT_HISTGRM_HISTORY.SAVTIME - 108379
SYS.WRI$_OPTSTAT_HISTGRM_HISTORY.SPARE6 - 108379
SYS.WRI$_OPTSTAT_HISTHEAD_HISTORY.SAVTIME - 74968
SYS.WRI$_OPTSTAT_HISTHEAD_HISTORY.SPARE6 - 74968
SYS.WRI$_OPTSTAT_IND_HISTORY.SAVTIME - 2739
SYS.WRI$_OPTSTAT_IND_HISTORY.SPARE6 - 2739
SYS.WRI$_OPTSTAT_OPR.END_TIME - 372
SYS.WRI$_OPTSTAT_OPR.SPARE6 - 372
SYS.WRI$_OPTSTAT_OPR.START_TIME - 372
SYS.WRI$_OPTSTAT_OPR_TASKS.END_TIME - 7924
SYS.WRI$_OPTSTAT_OPR_TASKS.SPARE6 - 7924
SYS.WRI$_OPTSTAT_OPR_TASKS.START_TIME - 7924
SYS.WRI$_OPTSTAT_TAB_HISTORY.SAVTIME - 4896
SYS.WRI$_OPTSTAT_TAB_HISTORY.SPARE6 - 4896
SYS.WRM$_DATABASE_INSTANCE.STARTUP_TIME_TZ - 7
SYS.WRM$_PDB_INSTANCE.OPEN_TIME_TZ - 3
SYS.WRM$_PDB_INSTANCE.STARTUP_TIME_TZ - 3
SYS.WRM$_PDB_IN_SNAP.OPEN_TIME_TZ - 23
SYS.WRM$_SNAPSHOT.BEGIN_INTERVAL_TIME_TZ - 23
SYS.WRM$_SNAPSHOT.END_INTERVAL_TIME_TZ - 23
SYS.XS$PRIN.END_DATE - 15
SYS.XS$PRIN.START_DATE - 15
Total count * of SYS TSTZ columns is : 416617
There are in total 162 SYS TSTZ columns.
.
For non-SYS tables ...
Note: empty tables are not listed.
Owner.TableName.ColumnName - COUNT(*) of that column
GSMADMIN_INTERNAL.AQ$_CHANGE_LOG_QUEUE_TABLE_S.CREATION_TIME - 1
GSMADMIN_INTERNAL.AQ$_CHANGE_LOG_QUEUE_TABLE_S.DELETION_TIME - 1
GSMADMIN_INTERNAL.AQ$_CHANGE_LOG_QUEUE_TABLE_S.MODIFICATION_TIME - 1
WMSYS.AQ$_WM$EVENT_QUEUE_TABLE_S.CREATION_TIME - 1
WMSYS.AQ$_WM$EVENT_QUEUE_TABLE_S.DELETION_TIME - 1
WMSYS.AQ$_WM$EVENT_QUEUE_TABLE_S.MODIFICATION_TIME - 1
WMSYS.WM$WORKSPACES_TABLE$.CREATETIME - 1
WMSYS.WM$WORKSPACES_TABLE$.LAST_CHANGE - 1
Total count * of non-SYS TSTZ columns is : 8
There are in total 20 non-SYS TSTZ columns.
Total Minutes elapsed : 0

```

Session altered.

SQL>

5° Script de Post Upgrade – utltz_upg_check.sql

SQL> @?/rdbms/admin/utltz_upg_check.sql

Session altered.

INFO: Starting with RDBMS DST update preparation.
INFO: NO actual RDBMS DST update will be done by this script.
INFO: If an ERROR occurs the script will EXIT sqlplus.
INFO: Doing checks for known issues ...
INFO: Database version is 19.0.0.0 .
INFO: Database RDBMS DST version is DSTv31 .
INFO: No known issues detected.
INFO: Now detecting new RDBMS DST version.
A prepare window has been successfully started.
INFO: Newest RDBMS DST version detected is DSTv32 .
INFO: Next step is checking all TSTZ data.
INFO: It might take a while before any further output is seen ...
A prepare window has been successfully ended.
INFO: A newer RDBMS DST version than the one currently used is found.
INFO: Note that NO DST update was yet done.
INFO: Now run utltz_upg_apply.sql to do the actual RDBMS DST update.
INFO: Note that the utltz_upg_apply.sql script will
INFO: restart the database 2 times WITHOUT any confirmation or prompt.

Session altered.

6° Script de Post Upgrade – utltz_upg_apply.sql

```
SQL> @?/rdbms/admin/utltz_upg_apply.sql
```

Session altered.

INFO: If an ERROR occurs, the script will EXIT SQL*Plus.

INFO: The database RDBMS DST version will be updated to DSTv32 .

WARNING: This script will restart the database 2 times

WARNING: WITHOUT asking ANY confirmation.

WARNING: Hit control-c NOW if this is not intended.

INFO: Restarting the database in UPGRADE mode to start the DST upgrade.

Database closed.

Database dismounted.

ORACLE instance shut down.

ORACLE instance started.

Total System Global Area 2466249672 bytes

Fixed Size 8899528 bytes

Variable Size 654311424 bytes

Database Buffers 1795162112 bytes

Redo Buffers 7876608 bytes

Database mounted.

Database opened.

INFO: Starting the RDBMS DST upgrade.

INFO: Upgrading all SYS owned TSTZ data.

INFO: It might take time before any further output is seen ...

An upgrade window has been successfully started.

INFO: Restarting the database in NORMAL mode to upgrade non-SYS TSTZ data.

Database closed.

Database dismounted.

ORACLE instance shut down.

ORACLE instance started.

Total System Global Area 2466249672 bytes

Fixed Size 8899528 bytes

Variable Size 654311424 bytes

Database Buffers 1795162112 bytes

Redo Buffers 7876608 bytes

Database mounted.

Database opened.

INFO: Upgrading all non-SYS TSTZ data.

INFO: It might take time before any further output is seen ...

INFO: Do NOT start any application yet that uses TSTZ data!

INFO: Next is a list of all upgraded tables:

Table list: "GSMADMIN_INTERNAL"."AQ\$_CHANGE_LOG_QUEUE_TABLE_S"

Number of failures: 0

Table list: "GSMADMIN_INTERNAL"."AQ\$_CHANGE_LOG_QUEUE_TABLE_L"

Number of failures: 0

Table list: "MDSYS"."SDO_DIAG_MESSAGES_TABLE"

Number of failures: 0

Table list: "DVSYS"."AUDIT_TRAIL\$"

Number of failures: 0

Table list: "DVSYS"."SIMULATION_LOG\$"

Number of failures: 0

INFO: Total failures during update of TSTZ data: 0 .

An upgrade window has been successfully ended.

INFO: Your new Server RDBMS DST version is DSTv32 .

INFO: The RDBMS DST update is successfully finished.

INFO: Make sure to exit this SQL*Plus session.

INFO: Do not use it for timezone related selects.

Session altered.

```
SQL>
```

Editar o /etc/oratab:

```
[oracle@upgrade db]$ vi /etc/oratab
```

```
+ASM:/u01/app/19/grid:N      # line added by Agent
ORAEX:/u01/app/19/db:N       # line added by Agent
#ORAEX:/u01/app/18/db:N      # line added by Agent
```

Upgrade do SRVCTL

```
[oracle@upgrade db]$ /u01/app/19/db/bin/srvctl upgrade database -d ORAEX -oraclehome /u01/app/19/db/
```

NOTA: NESTE PONTO É VALIDO LEMBRAR PARA ONDE ESTÁ APONTADO O SPFILE, REVISÃO:

```
SQL> show parameter spfile
```

NAME	TYPE	VALUE
spfile	string	/u01/app/19/db/dbs/spfileORAEX.ora

```
[oracle@upgrade ~]$ srvctl modify database -d ORAEX -p /u01/app/19/db/dbs/spfileORAEX.ora
```

```
[oracle@upgrade ~]$ srvctl config database -d ORAEX
```

Database unique name: ORAEX

Database name: ORAEX

Oracle home: /u01/app/19/db/

Oracle user: oracle

Spfile: /u01/app/19/db/dbs/spfileORAEX.ora

Password file:

Domain:

Start options: open

Stop options: immediate

Database role: PRIMARY

Management policy: AUTOMATIC

Disk Groups: DATA_ORAEX

Services:

OSDBA group:

OSOPER group:

Database instance: ORAEX

Validação pós Upgrade:

```
SQL> select TZ_VERSION from registry$database;
```

```
TZ_VERSION
```

```
-----  
32
```

```
1 row selected.
```

```
SQL> SELECT * FROM v$timezone_file;
```

```
FILENAME          VERSION  CON_ID
```

```
-----  
timezlg_32.dat      32      0
```

```
1 row selected.
```

```
SQL> col HOST_NAME format a30
```

```
set lines 230
```

```
col COMP_NAME format a55
```

```
set pages 900
```

```
col DESCRIPTION format a65
```

```
col ACTION_TIME format a29
```

```
col COMMENTS format a40
```

```
col ACTION format a9
```

```
col NAMESPACE format a9
```

```
col ID format 99999999
```

```
SQL> select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv$instance, v$database;
```

```
INSTANCE_NAME  HOST_NAME          VERSION  STATUS  OPEN_MODE  DATABASE_ROLE  SWITCHOVER_STAT  
US
```

```
-----  
ORAEX          upgrade.localdomain 19.0.0.0.0  OPEN    READ WRITE  PRIMARY        NOT ALLOWED
```

```
1 row selected.
```

```
SQL> select comp_name, version, status, modified from dba_registry;
```

```
COMP_NAME          VERSION  STATUS  MODIFIED
```

```
-----  
Oracle Database Catalog Views      19.0.0.0.0  UPGRADED      09-MAR-2021 19:42:09  
Oracle Database Packages and Types 19.0.0.0.0  UPGRADED      09-MAR-2021 19:42:09  
Oracle Real Application Clusters    19.0.0.0.0  UPGRADED      09-MAR-2021 19:46:45  
JServer JAVA Virtual Machine        19.0.0.0.0  UPGRADED      09-MAR-2021 19:43:55  
Oracle XDK                          19.0.0.0.0  UPGRADED      09-MAR-2021 19:44:50  
Oracle Database Java Packages       19.0.0.0.0  UPGRADED      09-MAR-2021 19:45:02  
OLAP Analytic Workspace             19.0.0.0.0  UPGRADED      09-MAR-2021 19:45:19  
Oracle XML Database                 19.0.0.0.0  UPGRADED      09-MAR-2021 19:48:29  
Oracle Workspace Manager            19.0.0.0.0  UPGRADED      09-MAR-2021 19:46:45  
Oracle Text                        19.0.0.0.0  UPGRADED      09-MAR-2021 19:46:11  
Oracle Multimedia                   19.0.0.0.0  UPGRADED      09-MAR-2021 19:49:17  
Spatial                            19.0.0.0.0  UPGRADED      09-MAR-2021 19:53:58  
Oracle OLAP API                     19.0.0.0.0  UPGRADED      09-MAR-2021 19:54:20  
Oracle Label Security                19.0.0.0.0  UPGRADED      09-MAR-2021 19:45:25  
Oracle Database Vault                19.0.0.0.0  UPGRADED      09-MAR-2021 19:45:40
```

```
15 rows selected.
```

```
SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;
```

```
PATCH_ID ACTION  STATUS  ACTION_TIME  DESCRIPTION  
-----  
29517242 APPLY   SUCCESS    09-MAR-21 07:57:35.409862 PM Database Release Update : 19.3.0.0.190416 (29517242)
```

```
1 row selected.
```

Ajuste do db_profile e grid_profile:

```

[oracle@upgrade ~]$ cd /home/oracle/
[oracle@upgrade ~]$ mv db_profile db_profile_old
[oracle@upgrade ~]$ mv db_profile_19c db_profile

[oracle@upgrade ~]$ cat db_profile
export ORACLE_BASE=/u01/app/19
export ORACLE_HOME=/u01/app/19/db
export ORACLE_SID=ORAEX
export PATH=/u01/app/19/db/bin:/u01/app/19/db/bin:/usr/bin:/bin:/usr/sbin:/sbin
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"

[grid@upgrade ~]$ mv grid_profile grid_profile_old
[grid@upgrade ~]$ mv grid_profile_19c grid_profile

[grid@upgrade ~]$ cat grid_profile
export ORACLE_BASE=/u01/app/19
export ORACLE_HOME=/u01/app/19/grid
export ORACLE_SID=+ASM
export PATH=/u01/app/19/grid/bin:/usr/bin:/bin:/usr/sbin:/sbin

```

Teste de baixar o banco e o grid e subir novamente:

```

[oracle@upgrade ~]$ sqlplus / as sysdba

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>

[root@upgrade bin]# . /home/grid/grid_profile
[root@upgrade bin]# crsctl stop has
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'upgrade'
CRS-2673: Attempting to stop 'ora.ARCH_ORAEX.dg' on 'upgrade'
CRS-2673: Attempting to stop 'ora.DATA_AUX_ORAEX.dg' on 'upgrade'
CRS-2673: Attempting to stop 'ora.DATA_ORAEX.dg' on 'upgrade'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'upgrade'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'upgrade' succeeded
CRS-2677: Stop of 'ora.DATA_ORAEX.dg' on 'upgrade' succeeded
CRS-2677: Stop of 'ora.ARCH_ORAEX.dg' on 'upgrade' succeeded
CRS-2677: Stop of 'ora.DATA_AUX_ORAEX.dg' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.asm' on 'upgrade'
CRS-2677: Stop of 'ora.asm' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'upgrade'
CRS-2677: Stop of 'ora.evmd' on 'upgrade' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'upgrade'
CRS-2677: Stop of 'ora.cssd' on 'upgrade' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'upgrade' has completed
CRS-4133: Oracle High Availability Services has been stopped.

[root@upgrade OPatch]# shutdown -r now

```

Fiz o restart do host justamente pra ver se a camada do grid vai subir corretamente, visto que mexi no /etc/oratab:

```
[root@upgrade ~]# crsctl status res -t
```

Name	Target	State	Server	State details	

Local Resources					

ora.ARCH_ORAEX.dg	OFFLINE	OFFLINE	upgrade	STABLE	
ora.DATA_AUX_ORAEX.dg	OFFLINE	OFFLINE	upgrade	STABLE	
ora.DATA_ORAEX.dg	ONLINE	OFFLINE	upgrade	STABLE	
ora.LISTENER.lsnr	ONLINE	ONLINE	upgrade	STABLE	
ora.asm	ONLINE	OFFLINE	upgrade	Instance Shutdown,ST	
			ABLE		
ora.ons	OFFLINE	OFFLINE	upgrade	STABLE	

Cluster Resources					

ora.cssd	1	ONLINE	OFFLINE	upgrade	STARTING
ora.diskmon	1	OFFLINE	OFFLINE		STABLE
ora.evmd	1	ONLINE	INTERMEDIATE	upgrade	STABLE
ora.oraex.db	1	OFFLINE	OFFLINE		Instance Shutdown,ST
			ABLE		

Segundos depois:

```
[root@upgrade ~]# crsctl status res -t
```

Name	Target	State	Server	State details	
Local Resources					
ora.ARCH_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.DATA_AUX_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.DATA_ORAEX.dg	ONLINE	ONLINE	upgrade	STABLE	
ora.LISTENER.lsnr	ONLINE	ONLINE	upgrade	STABLE	
ora.asm	ONLINE	ONLINE	upgrade	Started,STABLE	
ora.ons	OFFLINE	OFFLINE	upgrade	STABLE	
Cluster Resources					
ora.cssd	1	ONLINE	ONLINE	upgrade	STABLE
ora.diskmon	1	OFFLINE	OFFLINE		STABLE
ora.evmd	1	ONLINE	INTERMEDIATE	upgrade	STABLE
ora.oraex.db	1	OFFLINE	OFFLINE		Instance Shutdown,ST
			ABLE		

Oráculo do usuário:


```
[oracle@upgrade ~]$ srvctl config database -d ORAEX
Database unique name: ORAEX
Database name: ORAEX
Oracle home: /u01/app/19/db/
Oracle user: oracle
Spfile: +DATA_ORAEX/ORAEX/PARAMETERFILE/spfile.266.1066427785
Password file:
Domain:
Start options: open
Stop options: immediate
Database role: PRIMARY
Management policy: AUTOMATIC
Disk Groups: DATA_ORAEX
Services:
OSDBA group:
OSOPER group:
Database instance: ORAEX
```

```
[oracle@upgrade ~]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Mar 10 00:04:04 2021
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Connected to an idle instance.
```

```
SQL> startup
ORACLE instance started.
```

```
Total System Global Area 2466249672 bytes
```

```
Fixed Size 8899528 bytes
```

```
Variable Size 654311424 bytes
```

```
Database Buffers 1795162112 bytes
```

```
Redo Buffers 7876608 bytes
```

```
Database mounted.
```

```
Database opened.
```

```
SQL>
```

```
SQL>
```

```
SQL>
```

```
SQL>
```

```
SQL> col HOST_NAME format a30
```

```
set lines 230
```

```
col COMP_NAME format a55
```

```
set pages 900
```

```
col DESCRIPTION format a65
```

```
col ACTION_TIME format a29
```

```
col COMMENTS format a40
```

```
col ACTION format a9
```

```
col NAMESPACE format a9
```

```
col ID format 99999999SQL> SQL> SQL> SQL> SQL> SQL> SQL> SQL>
```

```
SQL> select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv$instance, v$database;
```

INSTANCE_NAME	HOST_NAME	VERSION	STATUS	OPEN_MODE	DATABASE_ROLE	SWITCHOVER_STAT
---------------	-----------	---------	--------	-----------	---------------	-----------------

ORAEX	upgrade.localdomain	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED
-------	---------------------	------------	------	------------	---------	-------------

```
SQL> select comp_name, version, status, modified from dba_registry;
```

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Database Catalog Views	19.0.0.0.0	UPGRADED	09-MAR-2021 19:42:09
Oracle Database Packages and Types	19.0.0.0.0	UPGRADED	09-MAR-2021 19:42:09
Oracle Real Application Clusters	19.0.0.0.0	UPGRADED	09-MAR-2021 19:46:45
JServer JAVA Virtual Machine	19.0.0.0.0	UPGRADED	09-MAR-2021 19:43:55
Oracle XDK	19.0.0.0.0	UPGRADED	09-MAR-2021 19:44:50
Oracle Database Java Packages	19.0.0.0.0	UPGRADED	09-MAR-2021 19:45:02
OLAP Analytic Workspace	19.0.0.0.0	UPGRADED	09-MAR-2021 19:45:19
Oracle XML Database	19.0.0.0.0	UPGRADED	09-MAR-2021 19:48:29

Oracle Workspace Manager	19.0.0.0.0	UPGRADED	09-MAR-2021 19:46:45
Oracle Text	19.0.0.0.0	UPGRADED	09-MAR-2021 19:46:11
Oracle Multimedia	19.0.0.0.0	UPGRADED	09-MAR-2021 19:49:17
Spatial	19.0.0.0.0	UPGRADED	09-MAR-2021 19:53:58
Oracle OLAP API	19.0.0.0.0	UPGRADED	09-MAR-2021 19:54:20
Oracle Label Security	19.0.0.0.0	UPGRADED	09-MAR-2021 19:45:25
Oracle Database Vault	19.0.0.0.0	UPGRADED	09-MAR-2021 19:45:40

15 rows selected.

SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
29517242	APPLY	SUCCESS	09-MAR-21 07:57:35.409862 PM	Database Release Update : 19.3.0.0.190416 (29517242)

Aplicando o último PSU para 19c

Foi necessário atualizar os OPatchs novamente e aplicar o opatch:

```
[oracle@upgrade db]$ cp /u02/midia/p6880880_121010_Linux-x86-64.zip .
[oracle@upgrade db]$ mv OPatch OPatch_old
[oracle@upgrade db]$ unzip p6880880_121010_Linux-x86-64.zip
```

```
[root@upgrade grid]# . /home/grid/grid_profile
[root@upgrade grid]# cd $ORACLE_HOME/OPatch
[root@upgrade OPatch]# ./opatchauto apply /u02/midia/32226239
```

NOTA:

BUG1 – Na hora de baixar o banco, o opatch não conseguiu baixar automaticamente. Baixei com srvtcl (user oracle) e matei a sessão do opatch. Voltei com o patch resume e funcionou.

BUG2 – Datapatch não correu automaticamente. Segui a nota **Doc ID 2650624.1**) para seguir com o datapatch.

```
[oracle@upgrade OPatch]$ ./datapatch -verbose
SQL Patching tool version 19.10.0.0.0 Production on Wed Mar 10 17:18:59 2021
Copyright (c) 2012, 2020, Oracle. All rights reserved.
```

Log file for this invocation: /u01/app/19/cfgtoollogs/sqlpatch/sqlpatch_6864_2021_03_10_17_18_59/sqlpatch_invocation.log

```
Connecting to database...OK
Gathering database info...done
Bootstrapping registry and package to current versions...done
Determining current state...done
```

Current state of interim SQL patches:
No interim patches found

Current state of release update SQL patches:
Binary registry:
19.10.0.0.0 Release_Update 210108185017: Installed
SQL registry:
Applied 19.3.0.0.0 Release_Update 190410122720 successfully on 10-MAR-21 11:39:47.714515 AM

Adding patches to installation queue and performing prereq checks...done
Installation queue:
No interim patches need to be rolled back
Patch 32218454 (Database Release Update : 19.10.0.0.210119 (32218454)):
Apply from 19.3.0.0.0 Release_Update 190410122720 to 19.10.0.0.0 Release_Update 210108185017
No interim patches need to be applied

Installing patches...
Patch installation complete. Total patches installed: 1

```
Validating logfiles...done
Patch 32218454 apply: SUCCESS
logfile: /u01/app/19/cfgtoollogs/sqlpatch/32218454/24018797/32218454_apply_ORAEX_2021Mar10_17_19_20.log (no errors)
SQL Patching tool complete on Wed Mar 10 17:26:25 2021
```

```
SQL> @$ORACLE_HOME/rdbms/admin/utlrp.sql
```

Session altered.

TIMESTAMP

```
-----
COMP_TIMESTAMP UTLRP_BGN          2021-03-10 17:41:24
```

```
DOC> The following PL/SQL block invokes UTL_RECOMP to recompile invalid
DOC> objects in the database. Recompilation time is proportional to the
DOC> number of invalid objects in the database, so this command may take
DOC> a long time to execute on a database with a large number of invalid
DOC> objects.
DOC>
DOC> Use the following queries to track recompilation progress:
DOC>
DOC> 1. Query returning the number of invalid objects remaining. This
DOC>    number should decrease with time.
DOC>    SELECT COUNT(*) FROM obj$ WHERE status IN (4, 5, 6);
DOC>
DOC> 2. Query returning the number of objects compiled so far. This number
DOC>    should increase with time.
DOC>    SELECT COUNT(*) FROM UTL_RECOMP_COMPILED;
DOC>
DOC> This script automatically chooses serial or parallel recompilation
DOC> based on the number of CPUs available (parameter cpu_count) multiplied
DOC> by the number of threads per CPU (parameter parallel_threads_per_cpu).
DOC> On RAC, this number is added across all RAC nodes.
DOC>
DOC> UTL_RECOMP uses DBMS_SCHEDULER to create jobs for parallel
DOC> recompilation. Jobs are created without instance affinity so that they
DOC> can migrate across RAC nodes. Use the following queries to verify
DOC> whether UTL_RECOMP jobs are being created and run correctly:
DOC>
DOC> 1. Query showing jobs created by UTL_RECOMP
DOC>    SELECT job_name FROM dba_scheduler_jobs
DOC>       WHERE job_name like 'UTL_RECOMP_SLAVE_%';
DOC>
DOC> 2. Query showing UTL_RECOMP jobs that are running
DOC>    SELECT job_name FROM dba_scheduler_running_jobs
DOC>       WHERE job_name like 'UTL_RECOMP_SLAVE_%';
DOC>#
```

PL/SQL procedure successfully completed.

TIMESTAMP

```
-----
COMP_TIMESTAMP UTLRP_END          2021-03-10 17:45:52
```

```
DOC> The following query reports the number of invalid objects.
DOC>
DOC> If the number is higher than expected, please examine the error
DOC> messages reported with each object (using SHOW ERRORS) to see if they
DOC> point to system misconfiguration or resource constraints that must be
DOC> fixed before attempting to recompile these objects.
DOC>#
```

OBJECTS WITH ERRORS

```
-----
0
```

```
DOC> The following query reports the number of exceptions caught during
DOC> recompilation. If this number is non-zero, please query the error
DOC> messages in the table UTL_RECOMP_ERRORS to see if any of these errors
DOC> are due to misconfiguration or resource constraints that must be
```

DOC> fixed before objects can compile successfully.
 DOC> Note: Typical compilation errors (due to coding errors) are not
 DOC> logged into this table: they go into DBA_ERRORS instead.
 DOC>#

ERRORS DURING RECOMPILATION

0

Function created.

PL/SQL procedure successfully completed.

Function dropped.

PL/SQL procedure successfully completed.

SQL> set lines 180
 set pages 40000
 col owner for a30
 select owner, count(*) from dba_objects
 where status = 'INVALID'
 group by owner
 /SQL> SQL> SQL> 2 3 4

no rows selected

SQL> col HOST_NAME format a30
 set lines 230
 col COMP_NAME format a55
 set pages 900
 col DESCRIPTION format a65
 col ACTION_TIME format a29
 col COMMENTS format a40
 col ACTION format a9
 col NAMESPACE format a9
 col ID format 99999999

select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv\$instance, v\$database;

select comp_name, version, status, modified from dba_registry;

select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;SQL> SQL> SQL> SQL> SQL> SQL> SQL>
 SQL> SQL> SQL> SQL>

INSTANCE_NAME	HOST_NAME	VERSION	STATUS	OPEN_MODE	DATABASE_ROLE	SWITCHOVER_STAT
ORAEX	upgrade.localdomain	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED

ORAEX	upgrade.localdomain	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED
-------	---------------------	------------	------	------------	---------	-------------

SQL> SQL>

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Database Catalog Views	19.0.0.0.0	VALID	10-MAR-2021 17:45:53
Oracle Database Packages and Types	19.0.0.0.0	VALID	10-MAR-2021 17:45:53
Oracle Real Application Clusters	19.0.0.0.0	OPTION OFF	10-MAR-2021 11:29:28
JServer JAVA Virtual Machine	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Oracle XDK	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Oracle Database Java Packages	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
OLAP Analytic Workspace	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Oracle XML Database	19.0.0.0.0	VALID	10-MAR-2021 17:45:54
Oracle Workspace Manager	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Oracle Text	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Oracle Multimedia	19.0.0.0.0	VALID	10-MAR-2021 17:45:55
Spatial	19.0.0.0.0	VALID	10-MAR-2021 17:45:56
Oracle OLAP API	19.0.0.0.0	VALID	10-MAR-2021 17:45:56
Oracle Label Security	19.0.0.0.0	VALID	10-MAR-2021 17:45:56

15 rows selected.

SQL> SQL>

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
29517242	APPLY	SUCCESS	10-MAR-21 11.39.47.714515 AM	Database Release Update : 19.3.0.0.190416 (29517242)
32218454	APPLY	SUCCESS	10-MAR-21 05.26.25.380116 PM	Database Release Update : 19.10.0.0.210119 (32218454)

SQL>